BRAUER GROUPS OF GENERALISED MARKOFF SURFACES

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Abstract. Let k be a field of characteristic zero with $a, b, c \in k^\times$ and let $m \in k \setminus \{0, 4abc\}$. We compute the Brauer group of the generalised Markoff surface $ax^2+by^2+cz^2-xyz = m$. We also construct several counterexamples to the integral Hasse principle, which are explained by the Brauer–Manin obstruction.

1. Introduction

Markoff introduced the equation

$$x^2 + y^2 + z^2 = 3xyz$$

[Mar79], whose positive integral solutions form an orbit under the action of a discrete group [BU04, BGS16]. With the change of variables $(X, Y, Z) = (3x, 3y, 3z)$, the equation becomes $X^2 + Y^2 + Z^2 - XYZ = 0$, which defines a singular Markoff surface.

Let k be a field of characteristic zero with $a, b, c \in k^\times$ and let $m \in k \setminus \{0, 4abc\}$. We study the smooth projective cubic surface

$$X_m : \quad (ax^2 + by^2 + cz^2)w - xyz = mw^3 \subset \mathbf{P}^3_k \tag{1.1}$$

and its affine part, the generalised Markoff surface

$$U_m : \quad ax^2 + by^2 + cz^2 - xyz = m. \tag{1.2}$$

The boundary is the split triangle

$$\mathscr{D} = D_x + D_y + D_z, \qquad D_x = \{x = w = 0\}, \quad D_y = \{y = w = 0\}, \quad D_z = \{z = w = 0\}. \tag{1.3}$$

For ordinary Markoff surfaces $x^2+y^2+z^2-xyz = m$, Ghosh and Sarnak [GS22] proved that the integral Hasse principle holds for almost all integer parameters and exhibited infinitely many failures. Colliot-Thélène, Xu and the second author [CTWX20] and Loughran and Mitankin [LM21] computed Brauer groups and studied the integral Brauer–Manin obstruction. S. Chen [Che21] computed the algebraic Brauer group of coefficient twists $ax^2 + y^2 + z^2 - xyz = m$ and constructed examples with order-four algebraic Brauer classes. Dao [Dao24] studied Brauer groups and integral local–global principles for other Markoff-type cubic surfaces. For the general family (1.2), we compute both the Brauer group $\mathrm{Br}(U_m)$ and the unramified Brauer group $\mathrm{Br}(X_m)$.

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Put

$$\Delta = m - 4abc. \tag{1.4}$$

We begin with our computation on the unramified Brauer group, which is proved in Theorem 3.1.

Theorem 1.1. *Let k be a field of characteristic zero with $a, b, c \in k^\times$ and let $m \in k\backslash\{0, 4abc\}$. One has $\mathrm{Br}(X_m) = \mathrm{Br}_1(X_m)$ and*

$$\mathrm{Br}(X_m)/\mathrm{Br}(k) \simeq \begin{cases} 0, & \{a\Delta, b\Delta, c\Delta, m\Delta\} \cap k^{\times 2} \neq \varnothing, \\ (\mathbf{Z}/2)^2, & ab, ac, am \in k^{\times 2} \text{ and } a\Delta \notin k^{\times 2}, \\ \mathbf{Z}/2, & \text{otherwise.} \end{cases} \tag{1.5}$$

We next compute the algebraic part $\mathrm{Br}_1(U_m)/\mathrm{Br}(k)$. Write

$$\chi_1 = [a\Delta], \qquad \chi_2 = [b\Delta], \qquad \chi_3 = [c\Delta], \qquad \chi_4 = [m\Delta] \quad \text{in } k^\times/k^{\times 2}. \tag{1.6}$$

These four square classes determine the Galois action on $\mathrm{Pic}(\overline{U}_m)$, which gives the following classification.

Theorem 1.2. *Let k be a field of characteristic zero with $a, b, c \in k^\times$ and let $m \in k\backslash\{0, 4abc\}$. Let N be the number of nonzero characters in (1.6) and put $\rho = \dim_{\mathbf{F}_2}\langle\chi_1, \dots, \chi_4\rangle$. Then*

$$\mathrm{Br}_1(U_m)/\mathrm{Br}(k) \simeq \begin{cases} (\mathbf{Z}/2)^{N-\rho}, & N < 4, \\ (\mathbf{Z}/2)^{5-\rho}, & N = 4 \text{ and every relation has even weight,} \\ \mathbf{Z}/4 \oplus (\mathbf{Z}/2)^{3-\rho}, & N = 4 \text{ and an odd-weight relation exists.} \end{cases} \tag{1.7}$$

Here a relation is a vector $\epsilon = (\epsilon_1, \dots, \epsilon_4) \in \mathbf{F}_2^4$ such that

$$\epsilon_1\chi_1 + \epsilon_2\chi_2 + \epsilon_3\chi_3 + \epsilon_4\chi_4 = 0.$$

Its weight is defined by $\#\{i : \epsilon_i = 1\}$ and an odd-weight relation is one whose weight is odd.

For the transcendental part $\mathrm{Br}(U_m)/\mathrm{Br}_1(U_m)$ over a number field, put

$$M = \frac{m}{abc}, \qquad D = \frac{\Delta}{abc} = M - 4,$$

and choose square roots in a fixed algebraic closure of k. Define

$$K_1 = k(\sqrt{a\Delta}), \qquad K_2 = k(\sqrt{b\Delta}), \qquad K_3 = k(\sqrt{c\Delta}),$$
$$L = k(\sqrt{a\Delta}, \sqrt{b\Delta}, \sqrt{c\Delta}, \sqrt{m\Delta}), \qquad \eta = \frac{\sqrt{M} - \sqrt{D}}{2}.$$

Both square roots defining η belong to L, and $\eta \in L^\times$ because $M - D = 4$. For a number field F, let $w_F = \#\mu(F)$.

Theorem 1.3. *Let k be a number field with $a, b, c \in k^\times$ and let $m \in k \setminus \{0, 4abc\}$. With the notation above, put*

$$E = \gcd\left(w_k, \frac{w_{K_1}}{2}, \frac{w_{K_2}}{2}, \frac{w_{K_3}}{2}\right), \qquad h = [L(\sqrt[E]{\eta}) : L].$$

Then $h \mid E$ and

$$\frac{\mathrm{Br}(U_m)}{\mathrm{Br}_1(U_m)} \simeq \mathbf{Z}/n_0, \qquad n_0 = \frac{E}{h}. \tag{1.8}$$

The result is independent of the choices of roots.

Theorem 5.2.1 determines the extension of the transcendental part by the algebraic part. Together with the splitting of the constant subgroup $\mathrm{Br}(k)$, this gives the full Brauer group $\mathrm{Br}(U_m)$.

In particular, if k is a number field and has a real embedding (for example, $k = \mathbf{Q}$), then $\mu(k) = \{\pm 1\}$. Hence the transcendental part is either 0 or $\mathbf{Z}/2$. In this case, square conditions in k determine the full Brauer group and whether the extension splits.

Theorem 1.4. *Let k be a number field with a real embedding $v_0 : k \hookrightarrow \mathbf{R}$, let $a, b, c \in k^\times$, and suppose $m \neq 0, 4abc$. Set*

$$A = \mathrm{Br}_1(U_m)/\mathrm{Br}(k), \qquad B = \mathrm{Br}(U_m)/\mathrm{Br}(k),$$

where A is given by Theorem 1.2. If at least one of $-a\Delta, -b\Delta, -c\Delta$ is not a square in k, then $B = A$. Otherwise $-D \in k^{\times 2}$; choose $d \in k$ with $d^2 = -D$ and $v_0(d) > 0$, and put

$$p = 1 + \frac{d}{2}, \qquad q = 1 - \frac{d}{2}.$$

Then A is killed by 2, and

$$B \simeq \begin{cases} A \oplus \mathbf{Z}/2, & p \in k^{\times 2} \text{ or } q \in k^{\times 2}, \\ \mathbf{Z}/4 \oplus (\mathbf{Z}/2)^2, & -q \in k^{\times 2} \text{ and } p \notin k^{\times 2}, \\ A, & \text{otherwise.} \end{cases} \tag{1.9}$$

In the first two cases, $B/A \simeq \mathbf{Z}/2$; its extension by A is split in the first case and nonsplit in the second.

We also apply these computations to integral points. In Section 7, we construct two explicit families over $\mathbf{Q}$ with real points and $\mathbf{Z}_p$-points for every prime p, but no integral points. For each case, a quaternion class gives a Brauer–Manin obstruction. The constructions follow the reciprocity approach of [GS22, CTWX20, Che21].

The paper is organized as follows. Section 2 develops the geometric and descent preliminaries, and Section 3 computes the unramified Brauer group. Section 4 computes the Picard lattice and the algebraic part, while Section 5 determines the full affine Brauer group. Section 6 specializes the results to number fields with a real embedding and to ordinary Markoff surfaces. Finally, Section 7 treats the integral Hasse failures.

Notation and conventions. Fix an algebraic closure $\overline{k}$ of k and put $G_k = \mathrm{Gal}(\overline{k}/k)$. Cohomology of varieties means étale cohomology, and cohomology of fields and Galois modules means continuous Galois cohomology. We use the cohomological Brauer group $\mathrm{Br}(V) = H^2(V, \mathbf{G}_\mathrm{m})_\mathrm{tors}$ and write

$$\mathrm{Br}_1(V) = \ker[\mathrm{Br}(V) \longrightarrow \mathrm{Br}(\overline{V})].$$

The notation (f, g) denotes a quaternion class. Square classes are written additively, with $[q] = 0$ meaning $q \in k^{\times 2}$. For a torsion group A, $A[n]$ denotes its subgroup killed by n. The Tate twist $\mathbf{Z}/n(-1)$ means $\mathrm{Hom}(\mu_n, \mathbf{Z}/n)$. The constant Brauer groups inject in all cases considered here, so we identify $\mathrm{Br}(k)$ with its image. We refer to $\mathrm{Br}_1(V)/\mathrm{Br}(k)$ and $\mathrm{Br}(V)/\mathrm{Br}_1(V)$ as the algebraic and transcendental parts, respectively.

2. Geometry and descent

Lemma 2.1. *The surface X_m is smooth and geometrically rational, and $U_m(k) \neq \varnothing$. It follows that both $\mathrm{Br}(k) \to \mathrm{Br}(X_m)$ and $\mathrm{Br}(k) \to \mathrm{Br}(U_m)$ are split injections, and $\mathrm{Br}(\overline{X}_m) = 0$.*

Proof. On the boundary, the derivatives with respect to x, y, z are $-yz, -xz, -xy$. They vanish simultaneously only at the three vertices of the triangle, where the derivative with respect to w takes the values a, b, and c, respectively. Thus there are no singular points at infinity. In the affine chart, a singular point satisfies

$$2ax = yz, \qquad 2by = xz, \qquad 2cz = xy.$$

If a coordinate vanishes, all three vanish, which forces $m = 0$. Otherwise these equations give $x^2 = 4bc$, $y^2 = 4ac$, $z^2 = 4ab$, and $xyz = 8abc$, so the surface equation forces $m = 4abc$. Both possibilities are excluded.

Every smooth cubic surface over an algebraically closed field is rational; see, for example, [Poo17]. The boundary vertex $[1 : 0 : 0 : 0]$ is a k-point. A smooth cubic surface with a k-point is k-unirational [Poo17, Remark 9.4.11]. Since k is infinite, unirationality implies that $X_m(k)$ is Zariski dense in X_m; hence the nonempty open subset U_m has a k-point. Evaluation at these points splits the inclusion of the constant Brauer group. Finally, geometric rationality gives $\mathrm{Br}(\overline{X}_m) = 0$. □

Lemma 2.2. *Every invertible regular function on $\overline{U}_m$ is constant, and $\mathrm{Pic}(\overline{U}_m)$ is a free abelian group of rank four. For $V = X_m$ or U_m, one has*

$$\mathrm{Pic}(V) = \mathrm{Pic}(\overline{V})^{G_k}, \qquad \mathrm{Br}_1(V)/\mathrm{Br}(k) = H^1(k, \mathrm{Pic}(\overline{V})).$$

For every $n \geq 1$,

$$H^2(U_m, \mu_n)/H^2(k, \mu_n) \simeq H^2(\overline{U}_m, \mu_n)^{G_k}. \tag{2.1}$$

Proof. The assertions about invertible regular functions and the Picard group of $\overline{U}_m$ follow from the calculation in Section 4.1.

For the assertions about the Picard and algebraic Brauer groups, apply the Hochschild–Serre spectral sequence to $\mathbf{G}_\mathrm{m}$. A rational point makes the pullback $H^j(k, \mathbf{G}_\mathrm{m}) \to H^j(V, \mathbf{G}_\mathrm{m})$ injective in every degree, so the usual obstruction maps vanish. This gives the stated identifications. For (2.1), the Kummer sequence gives $H^1(\overline{U}_m, \mu_n) = 0$ because the invertible regular functions are constant and the geometric Picard group is torsion-free. The only possible differential out of $H^2(\overline{U}_m, \mu_n)^{G_k}$ is then the d_3 into $H^3(k, \mu_n)$; it vanishes because the rational point also splits this pullback. The edge sequence is (2.1). We use these standard descent and Kummer constructions in the form explained in [Mil80]. □

3. The Brauer group of the projective surface

Throughout this section put $\Delta = m - 4abc$ and let $t = x/w$. We compute classes unramified along all prime divisors, including the three lines at infinity.

Theorem 3.1. *The Brauer group of X_m, modulo constant classes, is*

$$\operatorname{Br}(X_m)/\operatorname{Br}(k) \cong \begin{cases} 0, & \text{if at least one of } a\Delta, b\Delta, c\Delta, m\Delta \text{ belongs to } k^{*2}, \\ (\mathbf{Z}/2)^2, & \text{if } ab, ac, am \in k^{*2} \text{ and } a\Delta \notin k^{*2}, \\ \mathbf{Z}/2, & \text{otherwise.} \end{cases}$$

In every nontrivial case the quaternion class

$$B = (t^2 - 4bc, b\Delta)$$

belongs to $\operatorname{Br}(X_m)$ *and is nonconstant. It generates this quotient whenever the quotient is cyclic. In the second case, choose $u, v \in k^*$ with $u^2 = bc$ and $v^2 = m/a$; these exist since $bc = (ab)(ac)/a^2$ and $m/a = (am)/a^2$ are squares. Then the classes*

$$B = (t^2 - 4bc, b\Delta), \qquad C = ((t - 2u)(t - v), b\Delta)$$

generate $\operatorname{Br}(X_m)/\operatorname{Br}(k)$.

We first describe the conic pencil and its splitting characters. Let L_x be the line $x = w = 0$ on X_m. The planes $x = tw$ through L_x, after removal of their fixed component L_x, define the conic-bundle morphism

$$f : X_m \longrightarrow \mathbf{P}^1_k.$$

On $w \neq 0$ this morphism is given by the function $t = x/w$. Its finite fibres are the residual conics

$$by^2 + cz^2 - tyz + (at^2 - m)w^2 = 0. \tag{3.1}$$

The fibre at infinity is the union of the two k-lines $y = w = 0$ and $z = w = 0$. In particular, this fibre is split. The two quadratics

$$F_1(t) = t^2 - 4bc, \qquad F_2(t) = t^2 - m/a$$

are separable and have disjoint zero sets. Their four geometric zeros correspond to all the finite degenerate fibres of the pencil.

Lemma 3.2. *Let Q be a closed point of $\mathbf{P}^1\backslash\{\infty\} = \mathbf{A}^1_k$. If $F_1(Q) = 0$, the two geometric components of $f^{-1}(Q)$ are defined over $k(Q)(\sqrt{b\Delta})$. If $F_2(Q) = 0$, they are defined over $k(Q)(\sqrt{a\Delta})$. The corresponding generic conic has Brauer class*

$$A = (F_1, b(m - at^2)) = (F_1, b\Delta) + (F_2, a\Delta) \quad \text{in } \operatorname{Br}(k(t)). \tag{3.2}$$

Proof. Multiplying (3.1) by $4b$ and completing the square gives

$$(2by - tz)^2 = F_1 z^2 + 4b(m - at^2)w^2.$$

At a zero of F_1, this becomes $(2by - tz)^2 = 4b\Delta w^2$, proving the first assertion. At a zero of F_2, the equation is binary in y, z, and its discriminant is $F_1(Q) = \Delta/a$, whose square class is $a\Delta$. This establishes the second assertion and yields the first expression for A.

For the final identity, observe that $F_1 - F_2 = \Delta/a$. The quaternion identity

$$(r, r-s) = (r, -s) + (s, r-s)$$

follows from the Steinberg relation $(u, 1-u) = 0$ for $u \neq 0, 1$ (see [GS17, Example 1.3.4, p. 8]), with u being replaced by s/r. Taking $r = F_1$ and $s = \Delta/a$, and adding $(F_1, -ab)$ to both sides gives

$$\begin{aligned} A = (F_1, -abF_2) &= (F_1, (-ab)(-\Delta/a)) + (a\Delta, F_2) \\ &= (F_1, b\Delta) + (F_2, a\Delta), \end{aligned}$$

where $m - at^2 = -aF_2$. This uses the bilinearity and symmetry of quaternion symbols and proves (3.2). □

Let $\mathcal{N}$ be the set of closed points whose fibres are nonsplit degenerate. For $Q \in \mathcal{N}$ write

$$d_Q = \begin{cases} b\Delta, & F_1(Q) = 0, \\ a\Delta, & F_2(Q) = 0. \end{cases}$$

By definition, d_Q represents a nonzero quadratic character of $k(Q)$. Identify quadratic characters with square classes and set

$$R = \ker \left[\mathbf{F}_2^{\mathcal{N}} \longrightarrow k^*/k^{*2}, \quad (\epsilon_Q)_Q \longmapsto \prod_{Q \in \mathcal{N}} N_{k(Q)/k}(d_Q)^{\epsilon_Q} \right]. \tag{3.3}$$

Proposition 3.3. *If $\mathcal{N} \neq \varnothing$, there is an isomorphism*

$$\mathrm{Br}(X_m)/\mathrm{Br}(k) \cong R/\langle \mathbf{1} \rangle, \qquad \mathbf{1} = (1)_{Q \in \mathcal{N}}.$$

If $\mathcal{N} = \varnothing$, then $\mathrm{Br}(X_m) = \mathrm{Br}(k)$.

Proof. By [CTSk21, Lemma 11.3.2], contracting the split fibre components gives a smooth projective conic bundle $X' \to \mathbf{P}^1_k$ with reduced irreducible fibres and $\mathrm{Br}(X') = \mathrm{Br}(X_m)$. Its singular fibres are indexed by $\mathcal{N}$, with splitting characters d_Q; its fibre at infinity is smooth and split. If $\mathcal{N} \neq \varnothing$, the generic conic class A is ramified, and [CTSk21, Proposition 11.3.4] identifies $\mathrm{Br}(X_m)/\mathrm{Br}(k)$ with $R/\langle \mathbf{1} \rangle$, since the residue vector of A is $\mathbf{1}$. If $\mathcal{N} = \varnothing$, then A is constant and specializes to zero at infinity, hence $A = 0$; thus X' is k-rational and $\mathrm{Br}(X_m) = \mathrm{Br}(k)$.

Since $d_Q \in k^*$, [CTSk21, Corollary 11.3.5] gives representatives

$$\beta_\epsilon = \sum_{Q \in \mathcal{N}} \epsilon_Q(p_Q(t), d_Q), \qquad \epsilon \in R,$$

where p_Q is the monic irreducible polynomial of Q. Their pullbacks represent the classes in the quotient. □

Proof of Theorem 3.1. The fibres over the zeros of F_1 are split exactly when $b\Delta$ or $c\Delta$ is a square in k. Indeed, if bc is a square, both zeros are rational and the criterion is simply $b\Delta \in k^{*2}$. Otherwise their residue field is $k(\sqrt{bc})$, and a scalar from k^* becomes a square in this quadratic field exactly when its square class is 1 or bc. Similarly the fibres over the zeros of F_2 are split exactly when $a\Delta$ or $m\Delta$ is a square.

If the fibres in both pairs are split, then $\mathcal{N} = \varnothing$, so Proposition 3.3 gives a zero quotient. If the fibres in exactly one pair are split, then all nonsplit fibres belong to the other pair. That pair contributes either one quadratic closed point or two rational points with the same character; in either case $R = \langle \mathbf{1} \rangle$, so the quotient is again zero. The first case of the theorem follows.

Suppose now that neither pair is split. An irreducible quadratic contributes one coordinate to R and no corestriction condition, since $N_{k(Q)/k}(d_Q) = d_Q^2$. A rational pair contributes two coordinates and one condition given by their common nonzero character. If both quadratics are irreducible, R has dimension two; if exactly one splits into rational points, R again has dimension two. After quotienting by $\mathbf{1}$, the result in each case is $\mathbf{Z}/2$.

If both pairs are rational, the corestriction map has rank one when $[b\Delta] = [a\Delta]$ and rank two otherwise. Its kernel therefore has dimension three or two, respectively. Equality of these two nonzero characters means $ab \in k^{*2}$. Together with $bc, m/a \in k^{*2}$, this is equivalent to $ab, ac, am \in k^{*2}$, yielding the asserted rank-two case.

By [CTSk21, Corollary 11.3.5], B corresponds to the full residue vector on the first pair and zero on the second. When neither pair is split, this vector is neither 0 nor $\mathbf{1}$. Hence B has nonzero image and generates the quotient in the rank-one case. In the rank-two case, ordering the fibres as $2u, -2u, v, -v$, the vectors of B, C are $(1,1,0,0)$ and $(1,0,1,0)$, which form a basis modulo $\langle (1,1,1,1) \rangle$. The displayed classes extend and generate. □

Remark 3.4. Although the calculation uses the x-pencil, the result is symmetric in a, b, c. The case $(\mathbf{Z}/2)^2$ occurs exactly when a, b, c, m have the same square class and $a\Delta$ is nonsquare. In this case, choose compatible square roots of bc, ac, and ab and set

$$X = x/\sqrt{bc}, \qquad Y = y/\sqrt{ac}, \qquad Z = z/\sqrt{ab}.$$

Then

$$X^2 + Y^2 + Z^2 - XYZ = \frac{m}{abc},$$

so this case reduces over k to the usual Markoff calculation. For general coefficients, this change of variables need not be defined over k.

4. The Picard lattice and the algebraic Brauer group

Throughout this section, put $\Delta = m - 4abc$ and assume $abcm\Delta \neq 0$. For $q \in k^\times$, write $[q]$ for its class in $k^\times/k^{\times 2}$. For convenience, we identify this group with $H^1(k, \mathbf{Z}/2)$, and write its operation additively.

4.1. **The geometric configuration.** Choose $q_a, q_b, q_c, q_m \in \overline{k}$ with

$$q_a^2 = a\Delta, \qquad q_b^2 = b\Delta, \qquad q_c^2 = c\Delta, \qquad q_m^2 = m\Delta,$$

and put

$$r_1 = \frac{q_b q_c}{\Delta}, \qquad r_2 = \frac{q_a q_c}{\Delta}, \qquad r_3 = \frac{q_a q_b}{\Delta}.$$

Then $r_1^2 = bc$, $r_2^2 = ac$, $r_3^2 = ab$ and $r_1 r_2 r_3 = abc$. Set

$$X = x/r_1, \qquad Y = y/r_2, \qquad Z = z/r_3, \qquad M = m/(abc), \qquad D = \Delta/(abc). \tag{4.1}$$

Dividing the defining cubic by abc gives

$$(X^2+Y^2+Z^2)w - XYZ = Mw^3, \qquad D = M-4. \tag{4.2}$$

The diagonal projective automorphism $[x:y:z:w] \mapsto [x/r_1 : y/r_2 : z/r_3 : w]$ identifies $\overline{X}_m$ with the ordinary Markoff surface of parameter $M = m/(abc) \notin \{0,4\}$. It identifies their affine parts and carries the ordered boundary triangle to the corresponding triangle on the ordinary surface. We may therefore apply the geometric results of [CTWX20] after this change of coordinates.

Write $L_1 = D_x$, $L_2 = D_y$ and $L_3 = D_z$ for the boundary lines. The geometric Picard lattice of a smooth cubic surface is unimodular of rank seven.

The divisor sequence (see also [CTWX20, Proposition 2.2]) is

$$0 \longrightarrow \overline{k}[U_m]^\times/\overline{k}^\times \longrightarrow \bigoplus_{j=1}^{3} \mathbf{Z}[L_j] \longrightarrow \mathrm{Pic}(\overline{X}_m) \longrightarrow \mathrm{Pic}(\overline{U}_m) \longrightarrow 0.$$

The boundary intersection matrix has diagonal entries -1 and off-diagonal entries 1, with determinant 4. The boundary classes are therefore independent, and every invertible regular function on $\overline{U}_m$ is constant. It follows that

$$\overline{k}[U_m]^\times = \overline{k}^\times, \qquad \mathrm{Pic}(\overline{U}_m) = \mathrm{Pic}(\overline{X}_m)/\langle L_1, L_2, L_3\rangle. \tag{4.3}$$

By [CTWX20, Proposition 2.2], this quotient is torsion-free, hence free of rank four. In particular, the boundary span is primitive.

Proposition 4.1.1. *There is an isomorphism of Galois lattices*

$$\mathrm{Pic}(\overline{U}_m) \simeq \mathcal{L} := \{(z_1,z_2,z_3,z_4) \in \mathbf{Z}^4 : z_1+z_2+z_3+z_4 \equiv 0 \pmod 2\},$$

under which the Galois group acts diagonally through the quadratic characters

$$\chi_1 = [a\Delta], \quad \chi_2 = [b\Delta], \quad \chi_3 = [c\Delta], \quad \chi_4 = [m\Delta]. \tag{4.4}$$

Its splitting field is therefore

$$K = k(\sqrt{a\Delta}, \sqrt{b\Delta}, \sqrt{c\Delta}, \sqrt{m\Delta}) = k(\sqrt{ab}, \sqrt{ac}, \sqrt{am}, \sqrt{a\Delta}).$$

Proof. Let $B = \sum_j \mathbf{Z}[L_j]$, $\Lambda = B^\perp$, and let $F = L_2 + L_3$ be the fibre class of the conic pencil through L_1. In the original coordinates, choose one component in each of the four finite degenerate geometric fibres as follows:

$$\begin{aligned} C_\pm :&\quad \Delta x = \pm 2q_bq_cw, \quad q_by \mp q_cz = \Delta w, \\ Q_\pm :&\quad q_ax = \pm q_mw, \qquad\quad 2b\,q_ay = (\pm q_m + \Delta)z. \end{aligned} \tag{4.5}$$

In each case, replacing Δ by $-\Delta$ in the right-hand side of the second equation gives the complementary component, whose class is $F - C_\pm$ or $F - Q_\pm$, respectively. Define

$$\begin{aligned} &\varepsilon_1 = F - Q_+ - Q_-, \quad \varepsilon_2 = F - C_+ - C_-, \\ &\varepsilon_3 = C_- - C_+, \qquad\quad \varepsilon_4 = Q_- - Q_+. \end{aligned} \tag{4.6}$$

Each chosen component meets L_1 once and avoids L_2, L_3; components in distinct fibres are disjoint. Hence $\varepsilon_i \in \Lambda$ and $\varepsilon_i \cdot \varepsilon_j = -2\delta_{ij}$. Moreover,

$$\omega := F - C_+ - Q_+ = \tfrac{1}{2}(\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4) \in \Lambda. \tag{4.7}$$

The primitive boundary lattice B has determinant 4. Put $P = \mathrm{Pic}(\overline{X}_m)$ and $B^\vee = \mathrm{Hom}(B, \mathbf{Z})$. Primitivity of B means that P/B is torsion-free, so a basis of B extends to a basis of P. Every homomorphism $B \to \mathbf{Z}$ extends to $P \to \mathbf{Z}$. Unimodularity of P means that its intersection matrix has determinant ± 1; equivalently, intersection identifies P with $P^\vee = \mathrm{Hom}(P, \mathbf{Z})$. The composite $P \simeq P^\vee \to B^\vee$ is therefore surjective. It sends p to $(b \mapsto p \cdot b)$ and has kernel $B^\perp = \Lambda$. Hence $P/(B \oplus \Lambda) \simeq B^\vee/B$, so $[P : B \oplus \Lambda] = |\det B| = 4$. The determinant–index formula [Huy16, Chapter 14, (0.1)–(0.2)] gives

$$|\det B|\,|\det \Lambda| = [P : B \oplus \Lambda]^2 |\det P| = 16,$$

whence $|\det \Lambda| = 4$. Put $\Lambda_0 = \sum_i \mathbf{Z}\varepsilon_i$ and $\Lambda' = \Lambda_0 + \mathbf{Z}\omega \subseteq \Lambda$. The orthogonal basis ε_i gives $|\det \Lambda_0| = 16$. Since $2\omega \in \Lambda_0$ but $\omega \notin \Lambda_0$, we have $[\Lambda' : \Lambda_0] = 2$. The same determinant–index formula yields

$$|\det \Lambda'| = \frac{|\det \Lambda_0|}{[\Lambda' : \Lambda_0]^2} = \frac{16}{4} = 4 = |\det \Lambda|.$$

Applying it once more to the full-rank inclusion $\Lambda' \subseteq \Lambda$ gives $[\Lambda : \Lambda']^2 = |\det \Lambda'|/|\det \Lambda| = 1$. Therefore

$$\Lambda = \Lambda' = \sum_i \mathbf{Z}\varepsilon_i + \mathbf{Z}\omega.$$

Since Λ is primitive in the unimodular lattice P, every homomorphism $\Lambda \to \mathbf{Z}$ extends to P and is given by intersection with an element of P. The intersection map $P \to \Lambda^\vee := \mathrm{Hom}(\Lambda, \mathbf{Z})$ is therefore surjective. Its kernel is $P \cap (B \otimes \mathbf{Q}) = B$, since B is primitive. By (4.3), this gives a Galois-equivariant isomorphism

$$\mathrm{Pic}(\overline{U}_m) = P/B \xrightarrow{\sim} \Lambda^\vee, \qquad [p] \longmapsto (\lambda \mapsto p \cdot \lambda).$$

For $f \in \Lambda^\vee$, put $z_i = f(\varepsilon_i) \in \mathbf{Z}$. The relation $2\omega = \sum_i \varepsilon_i$ gives $2f(\omega) = \sum_i z_i$, so $\sum_i z_i$ is even. Conversely, any integral tuple with even sum defines a unique $f \in \Lambda^\vee$ by

$$f(\varepsilon_i) = z_i, \qquad f(\omega) = \tfrac{1}{2}\sum_i z_i,$$

since these values respect the sole relation among the generators of Λ. Hence $f \mapsto (z_i)$ identifies $\Lambda^\vee$ with the asserted lattice $\mathcal{L}$.

Let $\sigma_a, \sigma_b, \sigma_c, \sigma_m$ denote the formal sign changes of q_a, q_b, q_c, q_m, respectively, each fixing the other three symbols. These need not be automorphisms of K: the actual Galois action is obtained by restricting the sign action along

$$\mathrm{Gal}(K/k) \hookrightarrow \{\pm 1\}^4, \qquad g \longmapsto \left(\frac{g(q_a)}{q_a}, \frac{g(q_b)}{q_b}, \frac{g(q_c)}{q_c}, \frac{g(q_m)}{q_m}\right).$$

For example, if $q_a \in k^\times$, the first sign is always $+1$.

We compute the action of two of these formal sign changes, using the same notation for divisors and their classes. The equations give

$$\sigma_a(C_\pm) = C_\pm, \qquad \sigma_a(Q_\pm) = F - Q_\mp.$$

It follows that

$$\sigma_a(\varepsilon_1) = F - (F - Q_-) - (F - Q_+) = -\varepsilon_1,$$

while $\varepsilon_2, \varepsilon_3, \varepsilon_4$ are fixed. Likewise,

$$\sigma_m(C_\pm) = C_\pm, \qquad \sigma_m(Q_\pm) = Q_\mp,$$

so $\sigma_m(\varepsilon_4) = Q_+ - Q_- = -\varepsilon_4$, while the other three ε_i are fixed. The analogous computations show that σ_b and σ_c negate only ε_2 and ε_3, respectively. Restricting to the actual Galois group therefore gives exactly the characters in (4.4); the dual action has the same characters. Finally, adding the first character to the other three gives $[ab], [ac], [am]$, proving the equality of the splitting fields. □

Lemma 4.1.2. *In the notation of Proposition 4.1.1,*

$$\Lambda = \mathcal{L}^\vee = \mathbf{Z}^4 + \mathbf{Z}\omega, \qquad \omega = (1,1,1,1)/2.$$

An integral basis is

$$A_i = \omega - \varepsilon_i - \varepsilon_4 \quad (1 \le i \le 3), \qquad A_4 = -\varepsilon_4.$$

Proof. The first assertion was proved above. The coordinate matrix whose columns are $A_1, \dots, A_4$ relative to the integral basis $(\varepsilon_1, \varepsilon_2, \varepsilon_3, \omega)$ has determinant -1, so the A_i form an integral basis of Λ. □

4.2. **Computation of the algebraic part.** Let N denote the number of nonzero characters in (4.4), and let

$$d = \dim_{\mathbf{F}_2} \langle \chi_1, \chi_2, \chi_3, \chi_4 \rangle.$$

A relation has *odd weight* if it has the form $\sum_i \epsilon_i \chi_i = 0$ with $\sum_i \epsilon_i = 1$ in $\mathbf{F}_2$.

Theorem 4.2.1. *The algebraic part is given by*

$$\mathrm{Br}_1(U_m)/\mathrm{Br}(k) \simeq \begin{cases} (\mathbf{Z}/2)^{N-d}, & N < 4, \\ (\mathbf{Z}/2)^{5-d}, & N = 4 \text{ and no odd-weight relation exists}, \\ \mathbf{Z}/4 \oplus (\mathbf{Z}/2)^{3-d}, & N = 4 \text{ and an odd-weight relation exists}. \end{cases}$$

In the last case, $d \in \{2, 3\}$. If $N = 4$, an odd-weight relation exists precisely when at least one of

$$abc\Delta, \qquad abm\Delta, \qquad acm\Delta, \qquad bcm\Delta$$

is a square in k. Thus the possible algebraic part is isomorphic to one of 0, $\mathbf{Z}/2$, $(\mathbf{Z}/2)^2$, $(\mathbf{Z}/2)^3$, $(\mathbf{Z}/2)^4$, $\mathbf{Z}/4$ and $\mathbf{Z}/4 \oplus \mathbf{Z}/2$.

Proof. By Lemma 2.2 and Proposition 4.1.1, it suffices to compute $H^1(k,\mathcal{L})$. Put $\mathcal{M} = \bigoplus_i \mathbf{Z}(\chi_i)$. By the explicit description of $\mathcal{L}$ in Proposition 4.1.1, the coordinate-sum map modulo 2 gives the exact sequence of Galois modules

$$0 \longrightarrow \mathcal{L} \longrightarrow \mathcal{M} \longrightarrow \mathbf{Z}/2 \longrightarrow 0. \tag{4.8}$$

The last map sends (z_i) to $\sum_i z_i \bmod 2$; it is Galois-equivariant with trivial action on $\mathbf{Z}/2$, since sign changes disappear modulo 2. Here $H^1(k,\mathcal{M}) = (\mathbf{F}_2)^N$, and the map $\Phi : H^1(k,\mathcal{M}) \to H^1(k,\mathbf{Z}/2)$ sends the ith basis vector to χ_i. If $N < 4$, a trivial sign summand makes $\mathcal{M}^{G_k} \to \mathbf{Z}/2$ surjective. Exactness therefore identifies $H^1(k,\mathcal{L})$ with $\ker\Phi$. Since $\operatorname{im}\Phi = \langle\chi_1,\dots,\chi_4\rangle$ has dimension d, this kernel has dimension $N-d$ over $\mathbf{F}_2$. We obtain $H^1(k,\mathcal{L}) \simeq (\mathbf{Z}/2)^{N-d}$.

If $N = 4$, then $\mathcal{M}^{G_k} = 0$, giving

$$\begin{gathered} 0 \longrightarrow \mathbf{Z}/2\langle\kappa\rangle \longrightarrow H^1(k,\mathcal{L}) \longrightarrow R \longrightarrow 0, \\ R = \ker\big((\mathbf{F}_2)^4 \xrightarrow{(\chi_i)} H^1(k,\mathbf{Z}/2)\big). \end{gathered} \tag{4.9}$$

For $\epsilon = (\epsilon_i) \in R \subset (\mathbf{F}_2)^4$, choose its integer lift $\widetilde{\epsilon} = (\widetilde{\epsilon}_i) \in \mathcal{M}$ with $\widetilde{\epsilon}_i \in \{0,1\}$. Also represent each $\chi_i(g)$ by 0 or 1. The action on the ith coordinate is multiplication by $(-1)^{\chi_i(g)}$, so define

$$c_\epsilon(g) := \frac{g\widetilde{\epsilon} - \widetilde{\epsilon}}{2} = -(\widetilde{\epsilon}_i\chi_i(g))_{i=1}^4.$$

Here division by 2 is applied to an even integer vector. Since $\epsilon \in R$, the sum $\sum_i \widetilde{\epsilon}_i\chi_i(g)$ is even for every g; hence $c_\epsilon(g) \in \mathcal{L}$. Moreover, the displayed formula gives

$$c_\epsilon(gh) = c_\epsilon(g) + g\,c_\epsilon(h),$$

so c_ϵ is a cocycle. For each nonzero χ_i, the cocycle $g \mapsto -\chi_i(g)$ represents the generator of $H^1(k,\mathbf{Z}(\chi_i)) \simeq \mathbf{Z}/2$. The image of $[c_\epsilon]$ in $H^1(k,\mathcal{M}) \simeq (\mathbf{F}_2)^4$ is ϵ.

Finally, $2c_\epsilon(g) = g\widetilde{\epsilon} - \widetilde{\epsilon}$. It follows that $2[c_\epsilon]$ is the connecting image of the parity $\sum_i \widetilde{\epsilon}_i \bmod 2$. Since κ is the connecting image of $1 \in \mathbf{Z}/2$, we obtain

$$2[c_\epsilon] = \left(\sum_i \epsilon_i \bmod 2\right)\kappa. \tag{4.10}$$

The extension is therefore elementary abelian of rank $5-d$ if all relations have even weight; otherwise it is $\mathbf{Z}/4 \oplus (\mathbf{Z}/2)^{3-d}$.

An odd relation among four nonzero characters has weight three. Its three characters span a two-dimensional space, so $d \in \{2,3\}$. Their products give precisely the four stated square conditions. □

Remark 4.2.2. The proof gives generators as Galois cocycles, including in the order-four cases. For each pair of equal square classes among $\chi_1,\dots,\chi_4$, the following quaternion representatives adapt the construction of [CTWX20, Proposition 3.2 and Theorem 3.4] to coefficient twists. When $a = b = c = 1$, they recover the classes constructed there. In the

notation of (4.4), they are

$$\begin{array}{l|l} \chi_1 = \chi_2 & (z - 2\sqrt{ab}, a\Delta) \\ \chi_1 = \chi_3 & (y - 2\sqrt{ac}, a\Delta) \\ \chi_2 = \chi_3 & (x - 2\sqrt{bc}, b\Delta) \\ \chi_1 = \chi_4 & (x - \sqrt{m/a}, a\Delta) \\ \chi_2 = \chi_4 & (y - \sqrt{m/b}, b\Delta) \\ \chi_3 = \chi_4 & (z - \sqrt{m/c}, c\Delta). \end{array}$$

Under the indicated equality, each displayed square root belongs to k. If the second entry of a quaternion symbol is nonsquare, the two components of the corresponding zero divisor are conjugate over the associated quadratic extension, so the quaternion is unramified on U_m. If that entry is a square, the quaternion is trivial.

5. The full affine Brauer group

Throughout this section, $\Delta = m - 4abc \neq 0$ and $m \neq 0$. We describe the n-torsion of $\mathrm{Br}(U_m)/\mathrm{Br}(k)$ using finite Galois modules and recover the full Brauer group by adjoining the constant classes.

Retain the square roots q_a, q_b, q_c, q_m chosen in Section 4.1, so that

$$q_a^2 = a\Delta, \qquad q_b^2 = b\Delta, \qquad q_c^2 = c\Delta, \qquad q_m^2 = m\Delta,$$

and put

$$L = k(q_a, q_b, q_c, q_m), \quad M = \frac{m}{abc}, \quad D = \frac{\Delta}{abc}, \quad r = \frac{\Delta q_m}{q_a q_b q_c}, \quad s = \frac{\Delta^2}{q_a q_b q_c}, \quad \eta = \frac{r-s}{2}.$$

These definitions give $r^2 = M$, $s^2 = D$, $M - D = 4$, and

$$\eta^{-1} = \frac{r+s}{2}. \tag{5.1}$$

Denote the four sign characters by

$$\epsilon_i : G_k \longrightarrow \{1, -1\} \qquad (1 \leq i \leq 4).$$

They correspond to $a\Delta, b\Delta, c\Delta, m\Delta$, respectively.

5.1. **The boundary lattice and its period.** Let $\mathscr{D} = D_x + D_y + D_z = X_m \setminus U_m$ be the boundary triangle. Its three components and its three vertices are defined over k. Set

$$\Lambda = \{\lambda \in \mathrm{Pic}(\bar{X}_m) : \lambda \cdot D_x = \lambda \cdot D_y = \lambda \cdot D_z = 0\}.$$

The pair $(\bar{X}_m, \bar{\mathscr{D}})$ is a Looijenga pair: $\bar{X}_m$ is rational and $\bar{\mathscr{D}}$ is an anticanonical cycle of rational curves. We recall the algebraic definition of its period homomorphism from [GHK15, Definition 1.5].

Definition 5.1.1. The *period homomorphism* of $(\bar{X}_m, \bar{\mathscr{D}})$ is given by restriction of line bundles:

$$\phi : \Lambda \longrightarrow \mathrm{Pic}^0(\bar{\mathscr{D}}), \qquad [\mathcal{L}] \longmapsto [\mathcal{L}|_{\bar{\mathscr{D}}}].$$

Here $\mathrm{Pic}^0(\bar{\mathscr{D}})$ consists of line bundles having degree zero on each boundary component. The intersection conditions defining Λ ensure that restriction lands in this group. An orientation of the boundary cycle identifies this group with $\overline{k}^*$, as recalled below.

We recall the gluing description of [GHK15, Lemma 2.1]. Orient the boundary cycle as $D_x \to D_y \to D_z \to D_x$, and write $P_{ij} = D_i \cap D_j$ for its vertices. Each component is a projective line, so for $\mathcal{L} \in \mathrm{Pic}^0(\bar{\mathscr{D}})$ we can choose nowhere-vanishing sections σ_i of $\mathcal{L}|_{D_i}$. The product

$$\theta(\mathcal{L}) = \frac{\sigma_y(P_{xy})}{\sigma_x(P_{xy})}\,\frac{\sigma_z(P_{yz})}{\sigma_y(P_{yz})}\,\frac{\sigma_x(P_{zx})}{\sigma_z(P_{zx})}$$

identifies $\mathrm{Pic}^0(\bar{\mathscr{D}})$ with $\overline{k}^*$; the ratios are taken in the common fibre at each vertex. Multiplying σ_i by a nonzero constant leaves this product unchanged. Conversely, the three gluing scalars, modulo such changes of component trivializations, are determined by their product. Reversing the orientation inverts θ. We henceforth use θ to regard ϕ as a homomorphism $\Lambda \to \overline{k}^*$.

Every point of $D_x \setminus \{P_{xy}, P_{xz}\}$ is a smooth point of $\bar{\mathscr{D}}$ and hence defines a Cartier divisor on it. Therefore, any divisor $A = \sum_P n_P P$ supported in $D_x \setminus \{P_{xy}, P_{xz}\}$ is a Cartier divisor on the boundary. The notation $\mathcal{O}_{\bar{\mathscr{D}}}(A)$ denotes its associated line bundle, which is canonically trivial away from $\mathrm{Supp}(A)$, and in particular on D_y, D_z.

Lemma 5.1.2. *Use the coordinate $u = Y/Z$ on D_x, so that P_{xy} and P_{xz} have coordinates 0 and ∞. Let $A = \sum_P n_P P$ be a degree-zero Cartier divisor on $\bar{\mathscr{D}}$ supported in $D_x \setminus \{P_{xy}, P_{xz}\}$. For any rational function f on D_x with $\mathrm{div}_{D_x}(f) = A|_{D_x}$, one has*

$$\theta(\mathcal{O}_{\bar{\mathscr{D}}}(A)) = \frac{f(0)}{f(\infty)} = \prod_P u(P)^{n_P}.$$

Proof. Put $\mathcal{L} = \mathcal{O}_{\bar{\mathscr{D}}}(A)$ and use the component sections σ_i in the gluing formula above. Since $\mathrm{div}_{D_x}(f) = A|_{D_x}$, the divisor-sheaf description gives $\mathcal{L}|_{D_x} = f^{-1}\mathcal{O}_{D_x}$. On D_y, D_z, the line bundle is canonically trivial. We may therefore choose

$$\sigma_x = f^{-1}, \qquad \sigma_y = 1, \qquad \sigma_z = 1.$$

These are nowhere-vanishing sections of the respective line bundles. At the vertices, A has no support, so the sections can be compared using the common local trivialization 1. We obtain

$$\frac{\sigma_y(P_{xy})}{\sigma_x(P_{xy})} = f(0), \qquad \frac{\sigma_z(P_{yz})}{\sigma_y(P_{yz})} = 1, \qquad \frac{\sigma_x(P_{zx})}{\sigma_z(P_{zx})} = f(\infty)^{-1}.$$

The formula defining θ gives $\theta(\mathcal{L}) = f(0)/f(\infty)$. Taking

$$f(u) = \prod_P (u - u(P))^{n_P}$$

gives $f(\infty) = 1$ and $f(0) = \prod_P u(P)^{n_P}$, because $\sum_P n_P = 0$. Any other choice differs by a nonzero constant, which preserves the ratio. □

Proposition 5.1.3. *As a Galois lattice, Λ has the following coordinate model:*

$$\Lambda = \mathbb{Z}e_1 + \mathbb{Z}e_2 + \mathbb{Z}e_3 + \mathbb{Z}e_4 + \mathbb{Z}h, \qquad 2h = e_1 + e_2 + e_3 + e_4.$$

Here the e_i's are linearly independent in $\Lambda \otimes \mathbb{Q}$ with Galois action $g(e_i) = \epsilon_i(g)e_i$. The period is

$$\phi(e_1) = \phi(e_2) = \phi(e_3) = -1, \qquad \phi(e_4) = -\eta^2, \qquad \phi(h) = \eta. \tag{5.2}$$

Moreover,

$$\mathrm{Pic}(\bar{U}_m) \simeq \Lambda^\vee := \mathrm{Hom}(\Lambda, \mathbb{Z}).$$

Proof. Take $e_i = \varepsilon_i$ and $h = \omega$ from the proof of Proposition 4.1.1, which establishes the lattice structure, the Galois action and the duality assertion. It remains to apply Lemma 5.1.2 to the boundary intersections of their defining divisors.

We work in the normalized coordinates (4.1), retaining the notation for the transported components. Represent the fibre class $F = D_y + D_z$ by the smooth conic

$$F_0 : \quad X = 0, \qquad Y^2 + Z^2 = Mw^2.$$

Indeed, the plane sections $X = 0$ and $w = 0$ are $D_x + F_0$ and $D_x + D_y + D_z$ respectively, so $[F_0] = F$. Its two intersection coordinates on D_x satisfy $u^2 + 1 = 0$, so their product is 1. In these coordinates, the component equations (4.5) become

$$C_\pm : \quad X = \pm 2w, \quad Y \mp Z = sw,$$

$$Q_\pm : \quad X = \pm rw, \qquad Y = \frac{\pm r + s}{2} Z.$$

Setting $w = 0$ gives the u-coordinates $1, -1, (r+s)/2 = \eta^{-1}, (-r+s)/2 = -\eta$, respectively. All these curves meet the boundary only on D_x, transversely and away from the vertices. We obtain the following products of their intersection coordinates, counted with multiplicities:

E	F_0	C_+	C_-	Q_+	Q_-
$\prod_{P \in E \cap D_x} u(P)$	1	1	-1	η^{-1}	$-\eta$

For any of the surface divisors E defining ε_i, ω, with F represented by F_0, its support is disjoint from $D_y \cup D_z$. Hence $A := E|_{\bar{\mathscr{D}}}$ is a Cartier divisor supported in $D_x \setminus \{P_{xy}, P_{xz}\}$, and

$$\mathcal{O}_{\bar{X}_m}(E)|_{\bar{\mathscr{D}}} \simeq \mathcal{O}_{\bar{\mathscr{D}}}(A).$$

Since $[E] \in \Lambda$, this divisor A has degree zero. Lemma 5.1.2 therefore applies and gives, for example,

$$\phi(\omega) = \frac{1}{1 \cdot \eta^{-1}} = \eta, \qquad \phi(\varepsilon_4) = \frac{-\eta}{\eta^{-1}} = -\eta^2.$$

The same formula applied to $\varepsilon_1 = F - Q_+ - Q_-$, $\varepsilon_2 = F - C_+ - C_-$ and $\varepsilon_3 = C_- - C_+$ gives $\phi(\varepsilon_1) = \phi(\varepsilon_2) = \phi(\varepsilon_3) = -1$. With the generators defined by (4.6) and (4.7), and the identifications $e_i = \varepsilon_i$, $h = \omega$, these are precisely the period values in (5.2). □

The period homomorphism is Galois-equivariant: restriction to the boundary and the gluing identification $\mathrm{Pic}^0(\bar{\mathscr{D}}) \simeq \overline{k}^*$ are defined over k.

5.2. **A finite module computing the full group.** For a positive integer n, we define the finite Galois module

$$T_n = \frac{\{(\lambda, t) \in \Lambda \times \overline{k}^* : t^n = \phi(\lambda)\}}{\{(n\lambda, \phi(\lambda)) : \lambda \in \Lambda\}}. \tag{5.3}$$

The group law is addition in the first coordinate and multiplication in the second. It fits into an exact sequence

$$0 \longrightarrow \mu_n \longrightarrow T_n \longrightarrow \Lambda/n\Lambda \longrightarrow 0. \tag{5.4}$$

Every element is killed by n, and T_n is free of rank five over $\mathbb{Z}/n$. Indeed, multiplication by n sends (λ, t) to the defining relation $(n\lambda, \phi(\lambda))$; the quotient $\Lambda/n\Lambda$ is free of rank four, so (5.4) splits as a sequence of $\mathbb{Z}/n$-modules. Its Galois extension class is part of the calculation below.

Theorem 5.2.1. *The inclusion* $\mathrm{Br}(k) \to \mathrm{Br}(U_m)$ *is split by evaluation at a* k*-point of* U_m*. For every* $n \geq 1$*, there is an isomorphism*

$$(\mathrm{Br}(U_m)/\mathrm{Br}(k))\,[n] \simeq \frac{\mathrm{Hom}_{G_k}(T_n, \mathbb{Z}/n)}{(\Lambda^\vee)^{G_k}/n(\Lambda^\vee)^{G_k}}. \tag{5.5}$$

The denominator maps to the numerator by $\ell(\lambda, t) = \ell(\lambda) \bmod n$*. Taking the direct limit with respect to the maps induced by the Kummer sequences computes the full group modulo constants.*

For the direct limit appearing in Theorem 5.2.1, if $n \mid N$ and $d = N/n$, the reduction map $T_N \to T_n$ sends (λ, t) to (λ, t^d). The transition on the numerator sends a functional F to $dF \circ \mathrm{red}$ with values in $\mathbf{Z}/N$; on the denominator it sends $\ell \bmod n$ to $d\ell \bmod N$. These maps specify the direct limit, so evaluation at a rational point gives

$$\mathrm{Br}(U_m) \simeq \mathrm{Br}(k) \oplus \varinjlim_n \frac{\mathrm{Hom}_{G_k}(T_n, \mathbf{Z}/n)}{(\Lambda^\vee)^{G_k}/n(\Lambda^\vee)^{G_k}}.$$

We prove Theorem 5.2.1 using three lemmas. Fix $n \geq 1$, put $R = \mathbf{Z}/n$, and write $i : \bar{\mathscr{D}} \hookrightarrow \bar{X}_m$ and $j : \bar{U}_m \hookrightarrow \bar{X}_m$ for the inclusions. All geometric cohomology groups carry their natural G_k-action.

Lemma 5.2.2. *The multidegree map gives an exact sequence*

$$0 \longrightarrow \overline{k}^* \longrightarrow \mathrm{Pic}(\bar{\mathscr{D}}) \xrightarrow{\deg} \mathbf{Z}^3 \longrightarrow 0. \tag{5.6}$$

Its kernel is $\mathrm{Pic}^0(\bar{\mathscr{D}})$*, and* $\mathrm{Br}(\bar{\mathscr{D}}) = 0$*. Moreover,*

$$H^1(\bar{\mathscr{D}}, \mu_n) = \overline{k}^*[n] = \mu_n, \qquad H^2(\bar{\mathscr{D}}, \mu_n) = R^3. \tag{5.7}$$

The relative cohomology sequence of the pair $(\bar{X}_m, \bar{\mathscr{D}})$ *induces an exact sequence*

$$0 \longrightarrow \mu_n \longrightarrow H^2_c(\bar{U}_m, \mu_n) \longrightarrow \Lambda/n\Lambda \longrightarrow 0. \tag{5.8}$$

Proof. Let $\nu : \widetilde{\mathscr{D}} \to \bar{\mathscr{D}}$ be the normalization and let p run through the three nodes. The normalization sequence is

$$1 \longrightarrow \mathbf{G}_{m,\bar{\mathscr{D}}} \longrightarrow \nu_* \mathbf{G}_{m,\widetilde{\mathscr{D}}} \longrightarrow \bigoplus_p (i_p)_* \mathbf{G}_{m,p} \longrightarrow 1. \tag{5.9}$$

The relevant higher direct images of ν vanish on strict local branches. On invertible regular functions, the last map is $(c_x, c_y, c_z) \mapsto (c_y/c_x, c_z/c_y, c_x/c_z)$, with cokernel $\overline{k}^*$. Since $\mathrm{Pic}(\widetilde{\mathscr{D}}) = \mathbf{Z}^3$ and $\mathrm{Pic}(p) = \mathrm{Br}(\widetilde{\mathscr{D}}) = 0$, its cohomology gives (5.6) and $\mathrm{Br}(\bar{\mathscr{D}}) = 0$. The kernel of multidegree is the connected torus $\mathbf{G}_m$ of cyclic gluing parameters; since multidegree is locally constant, this kernel equals $\mathrm{Pic}^0(\bar{\mathscr{D}})$. The Kummer sequence and the divisibility of $\overline{k}^*$ now give (5.7).

Properness of $\bar{X}_m$ and the open–closed sequence $0 \to j_! \mu_n \to \mu_{n,\bar{X}_m} \to i_* \mu_{n,\bar{\mathscr{D}}} \to 0$ give

$$\begin{aligned} H^1(\bar{X}_m, \mu_n) &\longrightarrow H^1(\bar{\mathscr{D}}, \mu_n) \longrightarrow H^2_c(\bar{U}_m, \mu_n) \\ &\longrightarrow H^2(\bar{X}_m, \mu_n) \longrightarrow H^2(\bar{\mathscr{D}}, \mu_n). \end{aligned} \tag{5.10}$$

Since $\bar{X}_m$ is rational, $H^1(\bar{X}_m, \mu_n) = 0$ and $H^2(\bar{X}_m, \mu_n) = \mathrm{Pic}(\bar{X}_m)/n$. The last map is multidegree modulo n. By Proposition 4.1.1, the integral multidegree map $\mathrm{Pic}(\bar{X}_m) \to \mathbf{Z}^3$ is surjective with kernel Λ. Since $\mathbf{Z}^3$ is free, reduction modulo n gives an exact sequence

$$0 \longrightarrow \Lambda/n\Lambda \longrightarrow \mathrm{Pic}(\bar{X}_m)/n \longrightarrow (\mathbf{Z}/n)^3 \longrightarrow 0.$$

The last map in (5.10) therefore has kernel $\Lambda/n\Lambda$, proving (5.8). □

Lemma 5.2.3. *Let*

$$\mathcal{F} = \ker(\mathbf{G}_{m,\bar{X}_m} \longrightarrow i_* \mathbf{G}_{m,\bar{\mathscr{D}}}).$$

In degrees one and two, we have

$$H^1(\bar{X}_m, \mathcal{F}) = \ker(\phi : \Lambda \to \overline{k}^*), \qquad H^2(\bar{X}_m, \mathcal{F}) = \overline{k}^*/\phi(\Lambda).$$

For every $n \geq 1$, the relative Kummer sequence identifies (5.8) with (5.4) by a G_k-equivariant isomorphism

$$H^2_c(\bar{U}_m, \mu_n) \simeq T_n. \tag{5.11}$$

Proof. Write $X = \bar{X}_m$, $D = \bar{\mathscr{D}}$ and $U = \bar{U}_m$. The complexes below are complexes of discrete G_k-modules, with cohomological shifts $H^q(K[r]) = H^{q+r}(K)$. We write $\tau_{\leq r}$ for the canonical truncation preserving cohomology in degrees at most r.

Relative invertible regular functions and line bundles. Invertible regular functions lift locally across a closed immersion, so

$$1 \longrightarrow \mathcal{F} \longrightarrow \mathbf{G}_{m,X} \longrightarrow i_* \mathbf{G}_{m,D} \longrightarrow 1 \tag{5.12}$$

is exact. A class in $H^1(X, \mathcal{F})$ is represented by a line bundle $\mathcal{L}$ on X together with a trivialization $\alpha : \mathcal{L}|_D \xrightarrow{\sim} \mathcal{O}_D$. Indeed, local frames compatible with α have transition functions restricting to 1 on D, and conversely such transition functions glue a pair $(\mathcal{L}, \alpha)$. This is the relative Picard group $\mathrm{Pic}(X, D)$; see [BVS01, Section 2.1 and Proposition 2.5.1].

The groups of invertible regular functions on X and D are both $\overline{k}^*$, and restriction between them is the identity. The cohomology sequence of (5.12) therefore gives

$$H^1(X,\mathcal{F}) = \mathrm{Pic}(X,D) = \ker[\mathrm{Pic}(X) \to \mathrm{Pic}(D)] = \ker(\phi : \Lambda \to \overline{k}^*). \tag{5.13}$$

A line bundle trivial on D has multidegree zero there, so its class lies in Λ, and its restriction is trivial exactly when $\phi(\lambda) = 1$. Since $H^2(X, \mathbf{G}_m) = 0$, the same sequence identifies $H^2(X,\mathcal{F})$ with the cokernel of restriction on Picard groups. The integral multidegree map $\mathrm{Pic}(X) \to \mathbb{Z}^3$ is surjective. Thus every line bundle on D is represented, modulo restriction from X, by an element of $\mathrm{Pic}^0(D) = \overline{k}^*$; the remaining relations are precisely $\phi(\Lambda)$. Hence

$$H^2(X,\mathcal{F}) = \overline{k}^*/\phi(\Lambda).$$

The relative Kummer exact sequence. For $n \geq 1$, multiplication by n on $\mathcal{F}$ is surjective as a map of étale sheaves. On U this is ordinary Kummer. Near D, if $f|_D = 1$, the equation $T^n = f$ is étale and its restriction to D has the branch $T = 1$. Choosing this branch gives an étale-local root still restricting to 1 on D, including at the nodes. The kernel is μ_n on U and has zero stalks on D: an n-th root of unity whose restriction to D is 1 is itself 1 in the strict local neighbourhood. We therefore obtain

$$0 \longrightarrow j_!\mu_n \longrightarrow \mathcal{F} \xrightarrow{(\cdot)^n} \mathcal{F} \longrightarrow 0. \tag{5.14}$$

This is the relative Kummer sheaf sequence stated at the beginning of [BVS01, Section 2.5, p. 33]. Proposition 2.5.2 of [BVS01] describes its degree-one cohomology. Now we give the derived calculation needed in degree two.

A model for the relative derived global sections. Put $C = R\Gamma(X,\mathcal{F})$ and $K_Y = R\Gamma(Y, \mathbf{G}_m)$ for $Y = X, D$. By (5.12), $C \simeq \mathrm{Cone}(K_X \to K_D)[-1]$. Evaluation at a common rational boundary vertex splits off $(\overline{k}^*)[0]$ from both K_X and K_D, compatibly with restriction and the G_k-action. Let $\widetilde{K}_Y$ denote the complementary summands. Restriction is the identity on constant invertible regular functions, whose cone is acyclic, so

$$C \simeq \mathrm{Cone}(\widetilde{K}_X \to \widetilde{K}_D)[-1].$$

The reduced complexes have $H^0 = H^2 = 0$ and $H^1 = \mathrm{Pic}(Y)$, using the vanishing of both geometric Brauer groups. Hence $\tau_{\leq 2}\widetilde{K}_Y \simeq \mathrm{Pic}(Y)[-1]$. Taking the relative fibre and truncating in degrees at most two gives

$$\tau_{\leq 2} C \simeq [\mathrm{Pic}(X) \xrightarrow{\mathrm{res}} \mathrm{Pic}(D)] \quad \text{in degrees } 1, 2. \tag{5.15}$$

The subcomplex

$$C_\phi = [\Lambda \xrightarrow{\phi} \overline{k}^*] \quad \text{in degrees } 1, 2$$

embeds into this Picard complex. By the surjectivity of the integral multidegree map and (5.6), the quotient complex is $[\mathbb{Z}^3 \xrightarrow{\mathrm{id}} \mathbb{Z}^3]$, which is acyclic. We therefore have $\tau_{\leq 2} C \simeq C_\phi$ equivariantly, with the differential ϕ retained.

Applying Kummer and computing degree two. Apply $R\Gamma(X, -)$ to (5.14). The resulting distinguished triangle is

$$R\Gamma(X, j_!\mu_n) \longrightarrow C \xrightarrow{n} C \longrightarrow R\Gamma(X, j_!\mu_n)[1].$$

Therefore

$$R\Gamma(X, j_!\mu_n) \simeq \mathrm{Cone}(n : C \to C)[-1].$$

Properness of X identifies the left side with $R\Gamma_c(U, \mu_n)$. The long exact cohomology sequence shows that we may replace C by $\tau_{\leq 2}C \simeq C_\phi$ when computing the degree-two cohomology of this shifted cone. Hence

$$H^2_c(U, \mu_n) \simeq H^2\big(\mathrm{Cone}(n : C_\phi \to C_\phi)[-1]\big).$$

With compatible cone signs, the explicit complex on the right is

$$\Lambda \xrightarrow{d^1} \Lambda \oplus \overline{k}^* \xrightarrow{d^2} \overline{k}^* \quad \text{in degrees } 1, 2, 3, \tag{5.16}$$

where

$$d^1(\nu) = (n\nu, \phi(\nu)), \qquad d^2(\lambda, t) = \phi(\lambda)t^{-n}.$$

The composition is trivial because $\phi(n\nu) = \phi(\nu)^n$. Taking the kernel of d^2 modulo the image of d^1 gives

$$H^2_c(U, \mu_n) \simeq \frac{\{(\lambda, t) \in \Lambda \times \overline{k}^* : t^n = \phi(\lambda)\}}{\{(n\nu, \phi(\nu)) : \nu \in \Lambda\}} = T_n.$$

Here t represents a boundary line bundle $\mathcal{N}$ of multidegree zero with $\mathcal{N}^{\otimes n} \simeq \mathcal{L}|_D$, where $[\mathcal{L}] = \lambda$.

Every step is equivariant, and on representatives the action is $g[(\lambda, t)] = [(g\lambda, g(t))]$. The maps from the boundary sequence become

$$\mu_n \longrightarrow T_n, \quad \zeta \longmapsto [(0, \zeta)], \qquad T_n \longrightarrow \Lambda/n\Lambda, \quad [(\lambda, t)] \longmapsto \lambda \bmod n\Lambda.$$

They identify (5.8) with (5.4), including its Galois extension class, and prove (5.11). □

Lemma 5.2.4. *There is a G_k-equivariant isomorphism*

$$H^2(\bar{U}_m, \mu_n) \simeq \mathrm{Hom}(T_n, \mathbf{Z}/n). \tag{5.17}$$

Under this isomorphism and the identification $\mathrm{Pic}(\bar{U}_m) = \Lambda^\vee$, the first Chern class of ℓ is the functional

$$[(\lambda, t)] \longmapsto \ell(\lambda) \bmod n.$$

Passing to the quotient gives an induced equivariant isomorphism

$$\mathrm{Br}(\bar{U}_m)[n] \simeq \mathrm{Hom}(\mu_n, \mathbf{Z}/n)$$

under which the Kummer projection is restriction of functionals from T_n to μ_n.

Proof. Poincaré duality [Mil80] gives a perfect equivariant pairing

$$H^2(\bar{U}_m, \mu_n) \times H^2_c(\bar{U}_m, \mu_n) \longrightarrow H^4_c(\bar{U}_m, \mu_n^{\otimes 2}) \xrightarrow{\mathrm{tr}} \mathbf{Z}/n.$$

Here $\mathbf{Z}/n$ has trivial Galois action; the two coefficient twists have already combined to give the surface trace twist. Together with Lemma 5.2.3, this proves (5.17). Dualizing (5.4) gives

$$0 \longrightarrow \Lambda^\vee/n\Lambda^\vee \longrightarrow \mathrm{Hom}(T_n, \mathbf{Z}/n) \xrightarrow{\mathrm{res}} \mathrm{Hom}(\mu_n, \mathbf{Z}/n) \longrightarrow 0.$$

To identify the left map, lift a line bundle on $\bar{U}_m$ to a divisor class $\widetilde{\ell} \in \mathrm{Pic}(\bar{X}_m)$. Its image $\ell \in \Lambda^\vee$ is the functional $\lambda \mapsto \widetilde{\ell} \cdot \lambda$; changing the lift by boundary classes does not change this functional. The map forgetting support sends $[(\lambda, t)]$ to $\lambda \bmod n \in H^2(\bar{X}_m, \mu_n)$. Compatibility of cup products, cycle classes and the trace therefore shows that its pairing with the first Chern class of ℓ is $\widetilde{\ell} \cdot \lambda = \ell(\lambda) \bmod n$. The left map is therefore the first Chern class map.

To identify the right-hand map as well, compare this dual sequence with the geometric Kummer sequence

$$0 \longrightarrow \mathrm{Pic}(\bar{U}_m)/n \xrightarrow{c_1} H^2(\bar{U}_m, \mu_n) \longrightarrow \mathrm{Br}(\bar{U}_m)[n] \longrightarrow 0.$$

The middle terms are identified by duality, and the left terms by $\mathrm{Pic}(\bar{U}_m) = \Lambda^\vee$. The Chern class calculation above identifies the two left injections. Passing to their cokernels therefore gives the right-hand isomorphism and the commutative diagram

$$\begin{array}{ccc} H^2(\bar{U}_m, \mu_n) & \longrightarrow & \mathrm{Br}(\bar{U}_m)[n] \\ \sim\big\downarrow & & \big\downarrow\sim \\ \mathrm{Hom}(T_n, \mathbf{Z}/n) & \xrightarrow{F \mapsto F|_{\mu_n}} & \mathrm{Hom}(\mu_n, \mathbf{Z}/n). \end{array} \tag{5.18}$$

In particular, the bottom arrow is restriction because it is the dual of the inclusion $\mu_n \hookrightarrow T_n$.

Let F correspond by duality to the geometric restriction of a Kummer lift of $\beta \in \mathrm{Br}(U_m)[n]$. By naturality under base change, diagram (5.18) identifies the geometric restriction of β with $F|_{\mu_n}$. This is independent of the lift, since Picard functionals vanish on μ_n. □

Proof of Theorem 5.2.1. Choose $P_0 \in U_m(k)$ as in Lemma 2.1, and put

$$C_n = \ker\big[P_0^* : H^2(U_m, \mu_n) \longrightarrow H^2(k, \mu_n)\big].$$

Evaluation splits pullback from k, so Lemma 2.2 and Lemma 5.2.4 give

$$C_n \simeq H^2(\bar{U}_m, \mu_n)^{G_k} \simeq \mathrm{Hom}_{G_k}(T_n, \mathbf{Z}/n).$$

Moreover, Lemma 2.2 gives

$$\mathrm{Pic}(U_m) = (\Lambda^\vee)^{G_k}. \tag{5.19}$$

First Chern classes lie in C_n, because $\mathrm{Pic}(\mathrm{Spec}\, k) = 0$. By Lemma 5.2.4, the Chern class of ℓ corresponds to $(\lambda, t) \mapsto \ell(\lambda) \bmod n$.

Compatibility of the Kummer sequence with evaluation at P_0 therefore gives

$$0 \longrightarrow \mathrm{Pic}(U_m)/n \longrightarrow C_n \longrightarrow \ker\big[\mathrm{Br}(U_m) \xrightarrow{P_0^*} \mathrm{Br}(k)\big][n] \longrightarrow 0.$$

Evaluation splits the entire constant Brauer subgroup, so its kernel is isomorphic to $\mathrm{Br}(U_m)/\mathrm{Br}(k)$. Substituting (5.19) proves (5.5). The transition maps follow from the functoriality of the relative Kummer sequence and the duality pairing.

□

5.3. **An explicit formula for the transcendental part.** We determine the transcendental part using the connecting homomorphism in Galois cohomology. Retain $M = m/(abc)$ and $D = \Delta/(abc)$, and recall

$$r = \frac{\Delta q_m}{q_a q_b q_c}, \qquad s = \frac{\Delta^2}{q_a q_b q_c}, \qquad \eta = \frac{r-s}{2}.$$

These definitions give $r^2 = M$, $s^2 = D$ and $rs = q_m/(abc)$. Retain also

$$L = k(\sqrt{a\Delta}, \sqrt{b\Delta}, \sqrt{c\Delta}, \sqrt{m\Delta}).$$

Put

$$K_1 = k(\sqrt{a\Delta}), \qquad K_2 = k(\sqrt{b\Delta}), \qquad K_3 = k(\sqrt{c\Delta}).$$

Here and below, $\mu_j \subset F$ means that F contains all jth roots of unity in the fixed algebraic closure.

We first recall the direct-limit form of (5.4). For $N = dn$, the injections

$$T_n \longrightarrow T_N, \qquad [(\lambda, t)] \longmapsto [(d\lambda, t)]$$

induce the usual inclusions on μ_n and multiplication by d on $\Lambda/n\Lambda$. Setting $\mathcal{T} = \varinjlim_n T_n$, we obtain the direct-limit sequence

$$0 \longrightarrow \mathbf{Q}/\mathbf{Z}(1) \longrightarrow \mathcal{T} \longrightarrow \Lambda \otimes \mathbf{Q}/\mathbf{Z} \longrightarrow 0. \tag{5.20}$$

Here $\mathbf{Q}/\mathbf{Z}(1) = \bigcup_n \mu_n$ and $T_n = \mathcal{T}[n]$, so taking n-torsion recovers (5.4).

For the dual sequence, dualize (5.4) at each finite level and then take direct limits. Put

$$V_n = \mathrm{Hom}(T_n, \mathbf{Z}/n), \qquad \mathcal{V} = \varinjlim_n V_n, \qquad \mathcal{A} = \Lambda^\vee \otimes \mathbf{Q}/\mathbf{Z}.$$

Lemma 5.2.4 therefore gives

$$0 \longrightarrow \mathcal{A} \longrightarrow \mathcal{V} \xrightarrow{\mathrm{res}} \mathbf{Q}/\mathbf{Z}(-1) \longrightarrow 0. \tag{5.21}$$

Here $V_n = \mathcal{V}[n]$, and res is induced by restriction to μ_n. In this sense, $\mathcal{V}$ is the direct-limit version of the numerator. The injective transition maps give $\mathcal{V}^{G_k} = \varinjlim_n \mathrm{Hom}_{G_k}(T_n, \mathbf{Z}/n)$. Taking invariants in (5.21) and using Theorem 5.2.1 yields

$$I := \frac{\mathrm{Br}(U_m)}{\mathrm{Br}_1(U_m)} \simeq \ker\Big[(\mathbf{Q}/\mathbf{Z}(-1))^{G_k} \xrightarrow{\partial} H^1(k, \mathcal{A})\Big]. \tag{5.22}$$

For an invariant character t and any lift $f \in \mathcal{V}$,

$$\partial(t) = [g \longmapsto gf - f], \qquad (gf)(v) = f(g^{-1}v).$$

This class is independent of the lift: replacing f by $f + a$, with $a \in \mathcal{A}$, adds the coboundary $ga - a$. Using the direct limit allows us to consider a Brauer lift whose order exceeds that of its geometric image. The theorem below computes this kernel.

Theorem 5.3.1. *Let k be a field of characteristic zero with $a, b, c \in k^*$, and suppose $m \neq 0, 4abc$. Define*

$$\mathcal{I}_{a,b,c,m} = \left\{ n \geq 1 : \begin{array}{l} \mu_n \subset k, \\ \mu_{2n} \subset K_i \quad \text{for } i = 1,2,3, \\ \eta \in L^{\times n} \end{array} \right\}. \tag{5.23}$$

Then

$$\frac{\mathrm{Br}(U_m)}{\mathrm{Br}_1(U_m)} \simeq \varinjlim_{n \in \mathcal{I}_{a,b,c,m}} \mathbf{Z}/n, \tag{5.24}$$

where, for $n \mid n'$, the transition map sends $1 \bmod n$ to $n'/n \bmod n'$. The set $\mathcal{I}_{a,b,c,m}$ is closed under divisors and least common multiples, and is independent of the choices of square roots.

Proof. An invariant character of order n requires $\mu_n \subset k$. Under this assumption, choose a primitive nth root ζ_n and write $t_n = 1/n$ for the character $t_n(\zeta_n^j) = j/n$. Choose a primitive $2n$th root ξ_{2n}, an nth root $\sqrt[n]{\eta}$, and a lift $f \in \mathrm{Hom}(T_n, \frac{1}{n}\mathbf{Z}/\mathbf{Z})$ of t_n. Put $c(g) = gf - f$, so $[c] = \partial(t_n)$.

Suppose $\partial(t_n) = 0$, so $c(g) = ga - a$ for some $a \in \mathcal{A}$. For $i = 1, 2, 3$ and $g \in G_{K_i}$, the equality $g(e_i) = e_i$ gives $c(g)(e_i) = a(g^{-1}e_i) - a(e_i) = 0$. Since $\phi(e_i) = -1$ by (5.2), evaluation using the lift $[(e_i, \xi_{2n})]$ gives

$$\begin{aligned} c(g)(e_i) &= f([(e_i, g^{-1}(\xi_{2n}))]) - f([(e_i, \xi_{2n})]) \\ &= f([(0, g^{-1}(\xi_{2n})/\xi_{2n})]) \\ &= t_n\left(\frac{g^{-1}(\xi_{2n})}{\xi_{2n}}\right). \end{aligned} \tag{5.25}$$

The ratio belongs to μ_n because $\xi_{2n}^n = -1$. Faithfulness of t_n implies that every $g \in G_{K_i}$ fixes ξ_{2n}; hence $\mu_{2n} \subset K_i$. Likewise, G_L fixes h and $\phi(h) = \eta$, so evaluation using $[(h, \sqrt[n]{\eta})]$ gives

$$0 = c(g)(h) = t_n\left(\frac{g^{-1}(\sqrt[n]{\eta})}{\sqrt[n]{\eta}}\right) \qquad (g \in G_L).$$

Hence $\sqrt[n]{\eta} \in L$, and so $n \in \mathcal{I}_{a,b,c,m}$.

Conversely, assume $n \in \mathcal{I}_{a,b,c,m}$. Our chosen roots satisfy $\xi_{2n} \in K_1 \cap K_2 \cap K_3$ and $\sqrt[n]{\eta} \in L$. The lifts of the basis e_1, e_2, e_3, h, together with $\mu_n \subset k$, therefore show that G_L fixes T_n. Hence $c|_{G_L} = 0$, so $\partial(1/n)$ restricts to zero over L. Since G_L also fixes $\mathcal{A}$, inflation–restriction gives

$$0 \longrightarrow H^1(G, \mathcal{A}) \xrightarrow{\inf} H^1(k, \mathcal{A}) \xrightarrow{\mathrm{res}} H^1(L, \mathcal{A}), \qquad G = \mathrm{Gal}(L/k).$$

It follows that c descends to a cocycle on G, and it suffices to prove $[c] = 0$ in $H^1(G, \mathcal{A})$.

Put $K_4 = k(\sqrt{m\Delta})$, $H_i = \mathrm{Gal}(L/K_i)$ and $C_i = \mathbf{Q}/\mathbf{Z}(\chi_i)$ for $1 \leq i \leq 4$. By Proposition 4.1.1 and Lemma 4.1.2, $\Lambda_0 := \bigoplus_i \mathbf{Z}e_i \subset \Lambda$ has quotient $\mathbf{Z}/2$, generated

by h, with $2h = \sum_i e_i$. Dualizing with $\mathbf{Q}/\mathbf{Z}$ gives the exact sequence of G-modules

$$0 \longrightarrow \mathbf{Z}/2 \xrightarrow{\iota} \mathcal{A} \xrightarrow{\pi} \bigoplus_i C_i \longrightarrow 0, \qquad \pi(\ell) = (\ell(e_i))_i.$$

Here $\iota(1)(e_j) = 0$ and $\iota(1)(h) = 1/2$. Choose $\ell_i \in \mathcal{A}$ with $\ell_i(e_j) = \delta_{ij}/2$ and $\ell_i(h) = 1/4$. These values respect $2h = \sum_j e_j$. The same relation and $g(e_j) = (-1)^{\chi_j(g)} e_j$ give $g^{-1}h = h - \sum_j \chi_j(g) e_j$, hence

$$(g\ell_i - \ell_i)(e_j) = 0, \qquad (g\ell_i - \ell_i)(h) = -\frac{\chi_i(g)}{2} \quad \text{in } \mathbf{Q}/\mathbf{Z}.$$

We obtain $g\ell_i - \ell_i = \iota(\chi_i(g))$ and $\iota_*(\chi_i) = 0 \in H^1(G, \mathcal{A})$. Since L is the compositum of the quadratic extensions defined by the χ_i, these characters span $H^1(G, \mathbf{Z}/2) = \mathrm{Hom}(G, \mathbf{Z}/2)$. The cohomology sequence therefore gives an injection

$$H^1(G, \mathcal{A}) \hookrightarrow \prod_i H^1(G, C_i). \tag{5.26}$$

For each i, restriction $H^1(G, C_i) \to H^1(H_i, \mathbf{Q}/\mathbf{Z})$ is injective: if $\chi_i = 0$ it is the identity, and otherwise inflation–restriction and cyclic periodicity give

$$H^1(G/H_i, C_i) = \widehat{H}^{-1}(G/H_i, C_i) = \frac{\mathbf{Q}/\mathbf{Z}}{2(\mathbf{Q}/\mathbf{Z})} = 0.$$

For $i = 1, 2, 3$, the lift $[(e_i, \xi_{2n})]$ is H_i-fixed, so the computation (5.25) gives $\pi_{i,*}[c] = 0$.

For $i = 4$, it suffices to find $s_4 \in K_4^*$ with $s_4^n = \phi(e_4) = -\eta^2$: then $[(e_4, s_4)]$ is H_4-fixed and the same evaluation gives $\pi_{4,*}[c] = 0$. Since $rs = q_m/(abc)$, direct expansion gives

$$\eta^2 = \frac{r^2 + s^2 - 2rs}{4} = \frac{m + \Delta - 2q_m}{4abc} \in k(q_m) = K_4.$$

If $\xi_{2n} \in K_4$, then for every $g \in H_4$,

$$\alpha_g := \frac{g(\sqrt[n]{\eta})}{\sqrt[n]{\eta}} \in \mu_{2n} \subset K_4, \qquad \alpha_g^n = \frac{g(\eta)}{\eta} \in \{1, -1\}.$$

Since $g(\alpha_g) = \alpha_g$ and $g^2 = 1$ in G, we get

$$\alpha_g^2 = g(\alpha_g)\alpha_g = \frac{g^2(\sqrt[n]{\eta})}{\sqrt[n]{\eta}} = 1.$$

Thus $\left(\sqrt[n]{\eta}\right)^2$ is fixed by H_4 and belongs to $L^{H_4} = K_4$. We may take

$$s_4 := \xi_{2n}\left(\sqrt[n]{\eta}\right)^2 \in K_4^*, \qquad s_4^n = \xi_{2n}^n \eta^2 = -\eta^2.$$

If $\xi_{2n} \notin K_4$, then $\xi_{2n} \notin k$; since $\xi_{2n} \in K_i$ and $[K_i : k] \le 2$, $K_1 = K_2 = K_3 = k(\xi_{2n})$. It follows that $L = K_4(\xi_{2n})$ is quadratic over K_4. Its nontrivial involution τ negates q_a, q_b, q_c and fixes q_m. The definitions of r, s therefore give $\tau(r) = -r$ and $\tau(s) = -s$, hence $\tau(\eta) = -\eta$. We may therefore take

$$s_4 = N_{L/K_4}\left(\sqrt[n]{\eta}\right), \qquad s_4^n = N_{L/K_4}(\eta) = -\eta^2.$$

All four coordinate classes vanish, so (5.26) gives $[c] = 0$ in $H^1(G, \mathcal{A})$, and hence $\partial(1/n) = 0$.

We have shown that I contains an element of order n exactly when $n \in \mathfrak{I}_{a,b,c,m}$. The uniqueness of finite cyclic subgroups of $\mathbf{Q}/\mathbf{Z}$ yields (5.24) and implies closure under divisors and least common multiples. Changing the square roots replaces η by $\pm\eta^{\pm 1}$; since $\mu_{2n} \subset L$ implies $-1 \in L^{\times n}$, the criterion is independent of these choices. □

Corollary 5.3.2. *Let $k = \mathbf{R}$, let $a, b, c \in \mathbf{R}^*$, and suppose $m \neq 0, 4abc$. With $\Delta = m - 4abc$, one has*

$$\frac{\mathrm{Br}(U_m)}{\mathrm{Br}_1(U_m)} \simeq \begin{cases} \mathbf{Z}/2, & a\Delta < 0,\ b\Delta < 0,\ c\Delta < 0, \\ 0, & \text{otherwise.} \end{cases} \tag{5.27}$$

Proof. Since $\mu_n \subset \mathbf{R}$ only for $n = 1, 2$, Theorem 5.3.1 shows that only $n = 2$ can contribute a nontrivial quotient. For this order, $\mu_4 \subset K_i$ is equivalent to $K_i = \mathbf{C}$, hence to $a\Delta < 0$, $b\Delta < 0$, $c\Delta < 0$, respectively. If all three inequalities hold, then $L = \mathbf{C}$ and η is automatically a square in L. These sign conditions are therefore both necessary and sufficient. □

Corollary 5.3.3. *If k is a number field, we write $w_F = |\mu(F)|$ for each number field F, and put*

$$E = \gcd\left(w_k, \frac{w_{K_1}}{2}, \frac{w_{K_2}}{2}, \frac{w_{K_3}}{2}\right), \qquad h = [L(\sqrt[E]{\eta}) : L].$$

Then $h \mid E$ and

$$\boxed{\frac{\mathrm{Br}(U_m)}{\mathrm{Br}_1(U_m)} \simeq \mathbf{Z}/N, \qquad N = \frac{E}{h}.} \tag{5.28}$$

In particular, $N = \max \mathfrak{I}_{a,b,c,m}$ and $N \mid w_k$. If $E = 1$, the transcendental part is trivial. For $E = 2$ case, it is $\mathbf{Z}/2$ exactly when η is a square in L.

Proof. Over a number field, the cyclotomic conditions in Theorem 5.3.1 are exactly $n \mid E$: they say $n \mid w_k$ and $2n \mid w_{K_i}$ for every i. Choose u with $u^E = \eta$. Since $\mu_E \subset k \subset L$, the extension $L(u)/L$ is Galois, and

$$\mathrm{Gal}(L(u)/L) \hookrightarrow \mu_E, \qquad g \longmapsto g(u)/u$$

identifies $\mathrm{Gal}(L(u)/L)$ with μ_h, where $h = [L(u) : L] \mid E$. For every divisor n of E,

$$\eta \in L^{\times n} \iff u^{E/n} \in L \iff h \mid E/n.$$

The admissible indices in Theorem 5.3.1 are therefore precisely the divisors of E/h. Their largest element gives (5.28). □

Remark 5.3.4. For odd n, the condition $\mu_n \subset k$ implies $\mu_{2n} \subset k$. Hence

$$\mathrm{Br}(U_m)/\mathrm{Br}_1(U_m) \text{ contains a subgroup of order } n$$

if and only if

$$\mu_n \subset k \qquad \text{and} \qquad \eta \in L^{\times n}.$$

For odd n, the latter condition can be tested in $K_4 = k(\sqrt{m\Delta})$:

$$\eta \in L^{\times n} \iff \eta^2 \in K_4^{\times n}.$$

Consequently, for a number field k, the subgroup of elements of odd order in the transcendental part is cyclic of order

$$N_{\mathrm{odd}} = \max\{n \mid w_k : \ n \text{ odd and } \eta^2 \in K_4^{\times n}\}.$$

In particular, it is trivial when $\mu(k) = \{\pm 1\}$.

6. Consequences and examples

6.1. **Fields with a real embedding.** Let k be a number field with a real embedding $v_0 : k \hookrightarrow \mathbf{R}$ (for example, $k = \mathbf{Q}$). Every root of unity in k maps to a real root of unity, so $\mu(k) = \{\pm 1\}$ and $-1 \notin k^{\times 2}$. Theorem 5.3.1 therefore shows that $\mathrm{Br}(U_m)/\mathrm{Br}_1(U_m)$ is either 0 or $\mathbf{Z}/2$. If the transcendental part is nonzero, the cyclotomic conditions force

$$K_1 = K_2 = K_3 = k(\sqrt{-1}), \qquad -a\Delta, -b\Delta, -c\Delta \in k^{\times 2}.$$

These square conditions imply that ab, ac, bc are squares in k and that

$$-D = \frac{-a\Delta}{(ab)(ac)} \in k^{\times 2}.$$

Choose $d \in k$ with $d^2 = -D$ and $v_0(d) > 0$. The normalization (4.1) is then defined over k and identifies U_m with the ordinary Markoff surface

$$U_m \simeq_k \{X^2 + Y^2 + Z^2 - XYZ = 4 - d^2\}.$$

All square tests and the Galois-module calculation below take place over k. The real embedding ensures that $v_0(p) > 0$ for $p = 1 + d/2$, so $-p \notin k^{\times 2}$. This gives the following description of the full Brauer group over such number fields.

Theorem 6.1.1. *Let k be a number field with a real embedding $v_0 : k \hookrightarrow \mathbf{R}$, let $a, b, c \in k^*$, and let $m \in k \setminus \{0, 4abc\}$. Put $\Delta = m - 4abc$, $M = m/(abc)$, $D = M - 4$, and let $\chi_1, \dots, \chi_4$ be the characters of $a\Delta, b\Delta, c\Delta, m\Delta$. Set*

$$N = \#\{i : \chi_i \neq 0\}, \qquad \rho = \dim_{\mathbf{F}_2} \langle \chi_1, \dots, \chi_4 \rangle.$$

When $-a\Delta, -b\Delta, -c\Delta$ are all squares in k, we have $-D \in k^{\times 2}$; in this case choose $d \in k$ with $d^2 = -D$ and $v_0(d) > 0$, and put

$$p = 1 + \frac{d}{2}, \qquad q = 1 - \frac{d}{2}, \qquad M = 4pq.$$

Then the full Brauer group is as follows.

(1) *If at least one of $-a\Delta, -b\Delta, -c\Delta$ is not a square in k, or if all three are squares but none of $p, q, -q$ is a square in k, then*

$$\mathrm{Br}(U_m) = \mathrm{Br}_1(U_m) \simeq \mathrm{Br}(k) \oplus \begin{cases} (\mathbf{Z}/2)^{N-\rho}, & N < 4, \\ (\mathbf{Z}/2)^{5-\rho}, & N = 4, \text{ no odd-weight relation}, \\ \mathbf{Z}/4 \oplus (\mathbf{Z}/2)^{3-\rho}, & N = 4, \text{ an odd-weight relation}. \end{cases}$$

Here an odd-weight relation is a relation $\sum_i \epsilon_i \chi_i = 0$ with $\sum_i \epsilon_i = 1$ in $\mathbf{F}_2$.

(2) *Suppose* $-a\Delta, -b\Delta, -c\Delta$ *are all squares in* k *and at least one of* $p, q, -q$ *is a square in* k*. Then*

$$\operatorname{Br}(U_m)/\operatorname{Br}_1(U_m) \simeq \mathbf{Z}/2, \qquad \operatorname{Br}_1(U_m) \simeq \operatorname{Br}(k) \oplus (\mathbf{Z}/2)^\ell,$$

where

$$\ell = \begin{cases} 4, & M \in k^{\times 2}, \\ 2, & -M \in k^{\times 2}, \\ 3, & \text{otherwise.} \end{cases}$$

The full group is given by the following two cases:

(a) *If* $-q \in k^{\times 2}$ *and* $p \notin k^{\times 2}$*, then* $\ell = 3$ *and the extension is nonsplit, with*

$$\operatorname{Br}(U_m)/\operatorname{Br}(k) \simeq \mathbf{Z}/4 \oplus (\mathbf{Z}/2)^2. \tag{6.1a}$$

(b) *If* $p \in k^{\times 2}$ *or* $q \in k^{\times 2}$*, the extension splits, with*

$$\operatorname{Br}(U_m)/\operatorname{Br}(k) \simeq (\mathbf{Z}/2)^{\ell+1}. \tag{6.1b}$$

In all cases, the constant subgroup splits off:

$$\operatorname{Br}(U_m) \simeq \operatorname{Br}(k) \oplus \big(\operatorname{Br}(U_m)/\operatorname{Br}(k)\big).$$

Proof. Put $A = \operatorname{Br}_1(U_m)/\operatorname{Br}(k)$ and $B = \operatorname{Br}(U_m)/\operatorname{Br}(k)$. The formula for the algebraic part in (1) follows from Theorem 4.2.1. If one of $-a\Delta, -b\Delta, -c\Delta$ is not a square, Theorem 5.3.1 gives $B = A$. Assume henceforth that $-a\Delta, -b\Delta, -c\Delta$ are squares. The normalization (4.1) gives the ordinary Markoff surface with parameter $M = 4 - d^2$ and characters $[-1], [-1], [-1], [-M]$. Theorem 4.2.1 now gives $A = (\mathbf{Z}/2)^\ell$ as stated. Put

$$t = \sqrt{M}, \qquad F = k(t), \qquad L = F(i), \qquad \eta = \frac{t - id}{2}, \qquad \kappa = \frac{(1-i)(d+2) + (1+i)t}{4}.$$

Using $t^2 - (d+2)^2 = -2d(d+2)$, we complete the square:

$$\begin{aligned} p\eta = \frac{(d+2)(t-id)}{4} &= \frac{i}{8}(t^2 - (d+2)^2 - 2i(d+2)t) \\ &= \frac{i}{8}(t - i(d+2))^2 = \left(\frac{(1+i)(t - i(d+2))}{4}\right)^2 \\ &= \left(\frac{(1-i)(d+2) + (1+i)t}{4}\right)^2 = \kappa^2. \end{aligned}$$

Since $\kappa^2 = p\eta$, Theorem 5.3.1 gives

$$B/A \simeq \mathbf{Z}/2 \iff \eta \in L^{\times 2} \iff p \in L^{\times 2} \iff \{p, q, -q\} \cap k^{\times 2} \neq \varnothing.$$

For the last equivalence, the quadratic square test gives

$$\begin{aligned} p \in L^{\times 2} &\iff p \in k(i)^{\times 2} \text{ or } p/M \in k(i)^{\times 2} \\ &\iff p \in k(i)^{\times 2} \text{ or } q \in k(i)^{\times 2}, \end{aligned}$$

where the second line uses $p/M = 1/(4q)$. The first line follows by squaring $x + y\sqrt{M}$ and comparing coefficients: if $\sqrt{M} \notin k(i)$, the cross term forces $xy = 0$; otherwise the assertion is immediate. Likewise, a nonzero element of k is a square in $k(i)$ exactly when it or its negative is a square in k, since $(x + iy)^2 \in k$ with $x, y \in k$ forces $xy = 0$. As $v_0(p) > 0$, $-p \notin k^{\times 2}$, leaving precisely $p, q, -q$.

Suppose $u \in L$ and $u^2 = \eta$. The extension $0 \to A \to B \to \mathbf{Z}/2 \to 0$ splits exactly when the nonzero quotient class has a lift of order 2. By Theorem 5.2.1 and (5.18), this means finding $f \in \mathrm{Hom}_G(T_2, \mathbf{F}_2)$ with $f(\tau) = 1$, where $G = \mathrm{Gal}(L/k)$ and

$$\tau = [(0, -1)], \qquad v_j = [(e_j, i)] \ (1 \le j \le 3), \qquad v_h = [(h, u)].$$

These form a basis of T_2. Let ψ and θ be the characters of i and t, respectively, and put $E = e_1 + e_2 + e_3$. The sign characters of e_1, e_2, e_3, e_4 are $\psi, \psi, \psi, \psi + \theta$. Since $e_4 = 2h - E$, taking $\theta(g) \in \{0, 1\}$ gives

$$\begin{aligned} g(h) &= \frac{(-1)^{\psi(g)} E + (-1)^{\psi(g)+\theta(g)} e_4}{2} \\ &= (-1)^{\psi(g)+\theta(g)} h + (-1)^{\psi(g)} \theta(g) E \equiv h + \theta(g) E \pmod{2\Lambda}. \end{aligned}$$

Moreover, $g(\tau) = \tau$ and $g(v_j) = v_j$: when $g(i) = -i$, one has $[(-e_j, -i)] = [(e_j, i)]$ because $(2e_j, -1)$ is a defining relation in T_2; otherwise v_j is fixed directly. Under the projection $T_2 \to \Lambda/2\Lambda$ in (5.4), the difference $g(v_h) - v_h - \theta(g)(v_1 + v_2 + v_3)$ therefore maps to zero. The kernel of this projection is $\mu_2 = \langle \tau \rangle$, so

$$g(v_h) = v_h + \theta(g)(v_1 + v_2 + v_3) + c(g)\tau, \tag{6.2}$$

for a unique $c(g) \in \mathbf{F}_2$. Since τ, v_1, v_2, v_3 are fixed, we have

$$\begin{aligned} g_1 g_2(v_h) &= g_1\big(v_h + \theta(g_2)(v_1 + v_2 + v_3) + c(g_2)\tau\big) \\ &= g_1(v_h) + \theta(g_2)(v_1 + v_2 + v_3) + c(g_2)\tau \\ &= v_h + \big(\theta(g_1) + \theta(g_2)\big)(v_1 + v_2 + v_3) + \big(c(g_1) + c(g_2)\big)\tau. \end{aligned}$$

Comparing this with the formula for $g_1 g_2(v_h)$, and using $\theta(g_1 g_2) = \theta(g_1) + \theta(g_2)$, gives $c(g_1 g_2) = c(g_1) + c(g_2)$ by uniqueness of the coefficient of τ. It follows that $c : G \to \mathbf{F}_2$ is a character. Put $W = \langle \tau, v_1, v_2, v_3 \rangle$, a trivial G-module. Dualizing $0 \to W \to T_2 \to \mathbf{F}_2 \to 0$ and taking G-invariants gives

$$\mathrm{Hom}_G(T_2, \mathbf{F}_2) \longrightarrow \mathrm{Hom}(W, \mathbf{F}_2) \xrightarrow{\delta} H^1(G, \mathbf{F}_2), \qquad \delta(a, x_1, x_2, x_3) = a\, c + (x_1 + x_2 + x_3)\theta,$$

where $a = f(\tau)$ and $x_j = f(v_j)$. The formula for δ follows by applying f to $g(v_h) - v_h$ in (6.2). A functional with $a = 1$ therefore extends exactly when $c \in \langle \theta \rangle$.

To evaluate this obstruction, put $z = \kappa/u$, so $z^2 = p$. If $L \neq F = k(t)$, let σ generate $\mathrm{Gal}(L/F)$. Then $\sigma(t) = t$, $\sigma(i) = -i$, $\theta(\sigma) = 0$, and $\sigma(h) = -h$. By the definitions of v_h and c,

$$\begin{aligned} c(\sigma)\tau &= \sigma(v_h) - v_h \\ &= [(-2h, \sigma(u)/u)] \\ &= [(0, \eta\, \sigma(u)/u)] = [(0, \sigma(u)u)]. \end{aligned}$$

Here the third line adds the defining relation $(2h, \phi(h)) = (2h, \eta)$ from (5.3), and the last equality uses $u^2 = \eta$. Since $\tau = [(0,-1)]$ and $\mu_2 \hookrightarrow T_2$ is injective, this proves $(-1)^{c(\sigma)} = \sigma(u)u$. Direct expansion of the formula for κ gives

$$\sigma(\kappa)\kappa = \frac{(d+2+t)^2 + (t-d-2)^2}{16} = \frac{(d+2)^2 + t^2}{8} = \frac{d+2}{2} = p,$$

where $t^2 + d^2 = 4$. Finally, $u = \kappa/z$ and $z^2 = p$ give

$$(-1)^{c(\sigma)} = \sigma(u)u = \frac{\sigma(\kappa)\kappa}{\sigma(z)z} = \frac{p}{\sigma(z)z} = \frac{z}{\sigma(z)}.$$

As $\ker\theta = \mathrm{Gal}(L/F)$, this shows that $c \in \langle\theta\rangle$ exactly when $z \in F$, or equivalently $p \in F^{\times 2}$. If $L = F$, then $z \in F$, and $\ker\theta = 1$ implies $c \in \langle\theta\rangle$. By the square test in $F = k(\sqrt{M})$, the extension therefore splits exactly when p or q is a square in k, and then $B = A \oplus \mathbf{Z}/2$, giving (2b). If it does not split, the nontrivial quotient forces $-q$ to be a square and p not to be a square. It follows that $v_0(q) < 0$ and $v_0(M) < 0$, while $-M = 4p(-q)$ is not a square, so $\ell = 3$. Since $2A = 0$, the nonsplit extension is $B \simeq \mathbf{Z}/4 \oplus (\mathbf{Z}/2)^{\ell-1} = \mathbf{Z}/4 \oplus (\mathbf{Z}/2)^2$, giving (2a). If the quotient is trivial, $B = A$. Finally, the constants split by Theorem 5.2.1. □

6.2. **The ordinary Markoff specialization.** Suppose ab and ac are squares in k. Choose $r_1^2 = bc$, $r_2^2 = ac$ and $r_3 = r_1 r_2/c$, so that $r_3^2 = ab$ and $r_1 r_2 r_3 = abc$. The substitution

$$X = x/r_1, \qquad Y = y/r_2, \qquad Z = z/r_3$$

is defined over k and identifies U_m with the ordinary Markoff surface of parameter $M = m/(abc)$. Put $D = M - 4$ and $K = k(\sqrt{M}, \sqrt{D})$, and choose $\eta = (\sqrt{M} - \sqrt{D})/2$ as before. The algebraic part is

$$\mathrm{Br}_1(U_m)/\mathrm{Br}(k) \simeq \begin{cases} 0, & D \in k^{\times 2}, \\ (\mathbf{Z}/2)^2, & D \notin k^{\times 2},\ MD \in k^{\times 2}, \\ (\mathbf{Z}/2)^4, & D \notin k^{\times 2},\ M \in k^{\times 2}, \\ (\mathbf{Z}/2)^3, & [M], [D] \text{ are independent.} \end{cases} \tag{6.3}$$

This recovers the formulas of [CTWX20, Theorem 3.4].

For the transcendental part, define

$$\mathcal{I}_{\mathrm{ord}} = \{n \geq 1 : \mu_n \subset k \text{ and } -1, \eta \in K^{\times n}\}. \tag{6.4}$$

Theorem 4.8 of [CTWX20] gives

$$\mathrm{Br}(U_m)/\mathrm{Br}_1(U_m) \simeq \varinjlim_{n \in \mathcal{I}_{\mathrm{ord}}} \mathbf{Z}/n. \tag{6.5}$$

The transition maps are the natural embeddings in $\mathbf{Q}/\mathbf{Z}$. The sign difference between the root used there and our η does not change (6.4), because -1 is also required to be an n-th power. Over a number field, the group in (6.5) is finite cyclic. Formula (5.5) also determines the extension of this cyclic group by (6.3).

We verify explicitly that the conditions in (6.4) agree with those in Theorem 5.3.1. Put $F = k(\sqrt{D})$. The square-class assumptions give $K_1 = K_2 = K_3 = F$ and $L = K$. If

$\mu_{2n} \subset F$, then $-1 \in K^{\times n}$. Conversely, assume $\mu_n \subset k$ and $-1, \eta \in K^{\times n}$. If $K = F$, the power condition on -1 supplies a primitive $2n$-th root of unity in F (for odd n this already follows from $\mu_n \subset k$). If $[K : F] = 2$, the nontrivial automorphism changes the sign of $\sqrt{M}$ and fixes $\sqrt{D}$, so $N_{K/F}(\eta) = -1$. Taking the norm of an n-th root of η shows that $-1 \in F^{\times n}$ and gives the same cyclotomic containment. Hence the two index sets coincide.

Example 6.2.1. Let $k = \mathbf{Q}$. Theorem 6.1.1 gives

(1) For $U_{-12} : x^2 + y^2 + z^2 - xyz = -12$,

$$\mathrm{Br}_1(U_{-12}) \simeq \mathrm{Br}(\mathbf{Q}) \oplus (\mathbf{Z}/2)^3,$$
$$\mathrm{Br}(U_{-12}) \simeq \mathrm{Br}(\mathbf{Q}) \oplus \mathbf{Z}/4 \oplus (\mathbf{Z}/2)^2.$$

(2) For $U_7 : x^2 + y^2 + 4z^2 - xyz = 7$,

$$\mathrm{Br}_1(U_7) \simeq \mathrm{Br}(\mathbf{Q}) \oplus (\mathbf{Z}/2)^3,$$
$$\mathrm{Br}(U_7) \simeq \mathrm{Br}(\mathbf{Q}) \oplus (\mathbf{Z}/2)^4.$$

In both cases, the algebraic part is $(\mathbf{Z}/2)^3$ and the transcendental part is $\mathbf{Z}/2$. However, the extension is nonsplit in the first case but split in the second, which means the algebraic and transcendental parts alone do not determine the full Brauer group.

7. Integral Hasse failures over $\mathbf{Q}$

In this section, the coefficients and the parameter are integers. Let $\mathcal{U}_m \subset \mathbf{A}^3_{\mathbf{Z}}$ denote the integral model $ax^2 + by^2 + cz^2 - xyz = m$, with generic fibre U_m. Write

$$\mathcal{U}_m(\mathbb{A}_{\mathbf{Z}}) = U_m(\mathbf{R}) \times \prod_p \mathcal{U}_m(\mathbf{Z}_p).$$

Its integral Brauer–Manin set consists of tuples $(P_v)_v$ satisfying

$$\sum_v \mathrm{inv}_v\, B(P_v) = 0 \qquad \forall B \in \mathrm{Br}(U_m).$$

The integral Hasse failure means that this product of local sets is nonempty while $\mathcal{U}_m(\mathbf{Z})$ is empty. In the following two families, the failure is explained by a single quaternion class in $\mathrm{Br}(X_m)$. Local solubility follows by lifting smooth points modulo each prime to $\mathbf{Z}_p$, giving solutions to every congruence as well.

7.1. **A positive family.**

The norm identities used for ordinary Markoff surfaces in [GS22, CTWX20] and for their one-coefficient twists in [Che21] also give explicit families with three distinct coefficient square classes. We give the local calculations directly.

Proposition 7.1.1. *Let a, b, c be positive pairwise coprime odd integers satisfying*

$$a \equiv b \equiv c \equiv 1 \pmod 8, \qquad ac \equiv -1 \pmod 3.$$

Assume that every prime divisor of abc is congruent to ± 1 modulo 8. Let d be a positive odd integer coprime to abc, all of whose prime divisors are congruent to ± 1 modulo 8, and let $m = 4abc + 2bd^2$. Then

$$\mathcal{U}_m(\mathbb{A}_{\mathbf{Z}}) \neq \varnothing, \qquad \mathcal{U}_m(\mathbb{A}_{\mathbf{Z}})^{\mathrm{Br}(U_m)} = \varnothing.$$

In particular, the equation has no solution in $\mathbf{Z}^3$. The obstruction is given by $B = (x^2 - 4bc, 2)$.

Proof. Put $f = ax^2 + by^2 + cz^2 - xyz - m$. The identity

$$(2by - xz)^2 - 8b^2d^2 = (x^2 - 4bc)(z^2 - 4ab) \tag{7.1}$$

will be used to compute the local invariants. Since $b\Delta = 2b^2d^2$, Theorem 3.1 gives $B \in \mathrm{Br}(X_m) \subset \mathrm{Br}_1(U_m)$.

We first check local solubility. At 2, the point $(1, 1, 1)$ solves the equation modulo 2, and the derivative with respect to y is odd. At 3, we have $3 \nmid abcd$, and $m \equiv b(ac - 1) \not\equiv 0 \pmod 3$. The nonsingular binary form $by^2 + cz^2$ represents this nonzero residue, giving a smooth point with $x = 0$. A nonsingular binary quadratic form over an odd finite field represents every nonzero element: this follows either from its split form or from surjectivity of the norm from the quadratic field extension.

Let $p \geq 5$. If $p \mid a$, take $x = 0$ and represent $m \equiv 2bd^2 \neq 0$ by $by^2 + cz^2$; if $p \mid c$, take $z = 0$ and use $ax^2 + by^2$. Pairwise coprimality ensures these forms are nonsingular. If $p \mid b$, the point $(1, a + c, 1)$ solves the equation modulo p and has $\partial f/\partial y = -1$. For $p \nmid abc$, use $x = 0$ when $p \nmid m$. When $p \mid m$, choose $t \in \mathbf{F}_p^*$ with $t^2 \neq 4bc$; such a choice exists since at most two nonzero elements are excluded. The form $by^2 + cz^2 - tyz$ is nonsingular and represents $-at^2 \neq 0$, giving a smooth point with $x = t$. Hensel's lemma lifts all these points. A real point is $(0, \sqrt{m/b}, 0)$.

We now compute the invariants. At every odd prime dividing $abcd$, 2 is a square and B evaluates to zero. At any other odd prime for which 2 is not a square, (7.1) shows that $x^2 - 4bc$ cannot vanish modulo p, since b, d are units. Its value is therefore a unit and a norm from the unramified quadratic extension, and again the invariant is zero. The real invariant is zero as well.

At 2, we have $m \equiv 6 \pmod 8$. If x is odd, then $x^2 - 4bc \equiv 5 \pmod 8$. If x is even, the equation modulo 2 forces y and z to be both even or odd. The former is impossible modulo 4, and the equation modulo 8 then forces $x \equiv 4 \pmod 8$. It follows that

$$(x^2 - 4bc)/4 \equiv 3 \pmod 8.$$

Both cases give $\mathrm{inv}_2 B = 1/2$, by the formula $(u, 2)_2 = (-1)^{(u^2-1)/8}$ for odd units u. Hence the sum of invariants is always $1/2$. The integral Brauer–Manin set is therefore empty; reciprocity then rules out integral solutions. □

7.2. **A negative family of coefficient twists.** The negative family of Ghosh and Sarnak [GS22, Proposition 8.1(i)] admits the following coefficient twist. Its obstruction is detected by a class on the projective surface.

Proposition 7.2.1. *Let A be a positive odd integer and let v be a positive odd integer such that $v^2 > 2A$ and every prime divisor of v is congruent to 1 or 3 modulo 8. Put $m = 4A - 2v^2$ and consider the integral model*

$$\mathcal{U}_{A,v}: \quad Ax^2 + y^2 + z^2 - xyz = m \subset \mathbf{A}^3_{\mathbf{Z}},$$

with generic fibre $U_{A,v}$ and smooth projective completion $X_{A,v}$. Then $\mathcal{U}_{A,v}(\mathbf{Z}_p) \neq \varnothing$ for every prime p and $U_{A,v}(\mathbb{R}) \neq \varnothing$, but

$$\mathcal{U}_{A,v}(\mathbb{A}_{\mathbf{Z}})^{\mathrm{Br}(U_{A,v})} = \varnothing, \qquad \mathcal{U}_{A,v}(\mathbf{Z}) = \varnothing.$$

More precisely, the class

$$B = (x^2 - 4, -2) \in \mathrm{Br}(X_{A,v})$$

has invariant $1/2$ at every integral 2-adic point and invariant zero at all other integral local points, including the real place.

Proof. Here $\Delta = m - 4A = -2v^2$, so B is the quaternion class in $\mathrm{Br}(X_{A,v})$ given by Theorem 3.1. The conditions $m < 0$ and $\Delta \neq 0$ ensure that the surfaces are smooth.

We first establish local solubility. At 2, put $y = z = 1$ and solve

$$Ax^2 - x + 2 = m.$$

The value at $x = 0$ is even, while the derivative $2Ax - 1$ is odd. Hensel's lemma therefore gives a solution in $\mathbf{Z}_2$. At 3, we have $m \equiv A + v^2 \pmod 3$. If $3 \nmid m$, set $x = 0$ and represent the nonzero residue m by $y^2 + z^2$ over $\mathbf{F}_3$; the resulting point is smooth. If $3 \mid m$ and $A \equiv 2 \pmod 3$, the point $(1, 1, 0)$ solves the equation modulo 3 and the partial derivative about y is 2. The only other possibility with $3 \mid m$ is $3 \mid A$ and $3 \mid v$. Then $(2, 1, 1)$ is a solution modulo 3, and the partial derivative with respect to x, namely $4A - 1$, is a unit.

Let $p \geq 5$. If $p \nmid m$, a representation of m by $y^2 + z^2$ over $\mathbf{F}_p$, with $x = 0$, gives a smooth point. If $p \mid m$ and $p \nmid A$, put $x = 1$ and represent $-A$ by

$$y^2 + z^2 - yz$$

over $\mathbf{F}_p$. This nondegenerate binary quadratic form represents every nonzero element of $\mathbf{F}_p$, and the chosen point is smooth because its value is nonzero. If $p \mid m$ and $p \mid A$, the point $(2, 1, 1)$ is a solution modulo p, with derivative $4A - 1 \equiv -1 \pmod p$ in x. All these points lift by Hensel's lemma. Finally, the polynomial $(A + 2)t^2 - t^3$ takes the value $m < 0$ for some positive real t, yielding a real point.

For the invariant at 2, note that $m \equiv 2 \pmod 8$. If x is odd, then $x^2 - 4 \equiv 5 \pmod 8$ and $(x^2 - 4, -2)_2 = -1$. Suppose that x is even. Parity forces y, z to have the same parity. They cannot both be even: in that case the left side of the equation is divisible by 4, whereas $m \equiv 2 \pmod 4$. Hence y, z are odd. If $x \equiv 2 \pmod 4$, then $Ax^2 + y^2 + z^2 \equiv 6 \pmod 8$ and $xyz \equiv 2$ or $6 \pmod 8$, which is incompatible with $m \equiv 2 \pmod 8$. Hence $4 \mid x$. The equation then gives $xyz \equiv 0 \pmod 8$, so $8 \mid x$. It follows that

$$\frac{x^2 - 4}{4} \equiv -1 \pmod 8, \qquad (x^2 - 4, -2)_2 = (-1, -2)_2 = -1.$$

In both cases $\mathrm{inv}_2 B = 1/2$.

At 3, the element -2 is a square in $\mathbf{Q}_3$, so B vanishes. Let $p \neq 2, 3$. If -2 is a square in $\mathbf{Q}_p$, the same conclusion holds. Otherwise, no integral local point can have $x \equiv \pm 2 \pmod{p}$: substituting this value into the equation would give

$$(y \mp z)^2 \equiv -2v^2 \pmod{p},$$

contrary to the nonsquareness of -2. Here $p \nmid v$, since all prime divisors of v split in $\mathbf{Q}(\sqrt{-2})$ by hypothesis. It follows that $x^2 - 4$ is a p-adic unit. The Hilbert symbol of two units at an odd prime is trivial, so $\mathrm{inv}_p B = 0$.

At a real point we have $|x| > 2$. Indeed, if $|x| \leq 2$, then

$$m = Ax^2 + (y - xz/2)^2 + (1 - x^2/4)z^2 \geq 0,$$

contrary to $m < 0$. Hence $x^2 - 4 > 0$ and the real invariant vanishes. The sum of the local invariants of B is therefore always $1/2$. It follows that the integral Brauer–Manin set is empty and, by reciprocity, that no integral point exists. □

Example 7.2.2. Taking $A = 1$ in Proposition 7.2.1 recovers [GS22, Proposition 8.1(i)]. Taking $A = 3$ and $v = 3^n$ gives the integral Hasse failures

$$3x^2 + y^2 + z^2 - xyz = 12 - 2 \cdot 9^n \qquad (n \geq 1).$$

For $n = 1, 2$, the character tuple is $([-6], [-2], [-2], [3])$, with $N = 4$, $d = 2$ and the odd-weight relation $\chi_1 + \chi_2 + \chi_4 = 0$. Theorem 4.2.1 gives

$$\mathrm{Br}(U_{3,3^n})/\mathrm{Br}(\mathbf{Q}) \simeq \mathbf{Z}/4 \oplus \mathbf{Z}/2 \qquad (n = 1, 2).$$

This is the full group by Theorem 6.1.1, since the coefficient classes $[3], [1], [1]$ do not all agree.

Remark 7.2.3. Integral Hasse failures need not be explained by the Brauer–Manin obstruction. For the ordinary Markoff surface with $m = 342 = 4 + 2 \cdot 13^2$, Ghosh–Sarnak [GS22, Proposition 8.1(iii)] prove that there are no integral points, whereas Colliot-Thélène–Wei–Xu [CTWX20, Proposition 5.11] prove that the integral Brauer–Manin set is nonempty. More generally, the same two conclusions hold for $m = 4 + 2\ell^2$ when $\ell \geq 13$ is prime, $\ell \equiv \pm 4 \pmod{9}$ and $\ell \equiv \pm 3 \pmod{8}$. There are infinitely many such primes, for example the primes in the progression $\ell \equiv 13 \pmod{72}$. The proof of nonexistence uses reduction theory as well as reciprocity.

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